From Threshold Belief to Epistemic Phase Geometry: Mechanized Dynamics in a Four-Valued Probabilistic Epistemic Logic

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Abstract

I investigate how four-valued evidence changes under probability-only conditionalization and what is lost when quantitative support is reduced to categorical belief. In the 4-PEL semantics, belief thresholds positive and negative support separately. A primitive, qualitative operator called evidence-stable knowledge returns that belief value on homogeneous accessible profiles and strict falsity otherwise. The term "knowledge" denotes this stipulated operator, not an independently established theory of epistemic justification.

A six-world witness family realizes all sixteen source–target values of $\mathsf{K}(\mathsf{B}p)$. For a chosen rational affine interpolation of support endpoints, threshold-bit changes determine unique wall-hit parameters; their order determines adjacent intermediate states. These parameters describe the interpolation, not an inferred history of the discrete update. Separately, normalized nonnegative world weights yield certified finite measures and convex model-valued paths whose fixed-event masses are affine. Formula-level identification for probability-sensitive events and generation of intermediate models by conditionalization remain open.

A further six-cell refinement records reliability tags while preserving four-cell support and convex mixing. Fine modal stability is equivalent to coarse stability plus local fibre rigidity; explicit S5-frame counterexamples show that coarse stability alone loses reliability information. This is a semantic projection result, not an axiomatization of a six-valued logic.

At the coarsest evidence level, the bilateral positive/negative signature induces a two-class matroid closure: same-polarity atoms are parallel, closure preserves the four-valued state, and the channel ranks of $\mathbf{N}, \mathbf{T}, \mathbf{F}, \mathbf{B}$ are respectively $0, 1, 1, 2$. The interaction with the existing topological evidence semantics has an exact boundary: topological closure satisfies the repository's exchange axiom exactly when singleton specialization is symmetric, the R_0 condition for genuine topological spaces. The existing Sierpiński witness fails this boundary, while the coarse positive/negative channel matroid is recovered as the closure of its canonical partition topology.

The circuit structure of this coarse matroid is completely classified: minimal dependent evidence sets are exactly distinct same-polarity pairs. Gate 4 upgrades the closure presentation to an explicit finite-ground layer with genuine deletion and contraction minors. Deleting one member of a parallel pair preserves the coarse FDE state, whereas deleting a unique channel representative removes exactly that support direction. Contracting one atom makes precisely its remaining same-polarity partners loops and removes nonloop capacity from that channel while preserving any opposite independent channel. This separates polarity occupancy from residual independent-information capacity after minors, and also separates structural contraction from a memory-bearing epistemic acceptance update.

Gate 5 asks when the four-valued layer recovers classical behavior from evidence structure rather than by stipulation. For a realized FDE value, channel completeness is equivalent to gap-freedom, channel consistency to glut-freedom, and their conjunction isolates exactly $\{\mathbf{T}, \mathbf{F}\}$. The recovered slice is closed under the existing FDE negation, conjunction, and disjunction. Tolerant excluded middle requires only gap-freedom, LP ex falso only glut-freedom, while strict-$\mathbf{T}$ excluded middle characterizes the full classical slice. Modal stability is not needed for local bivalence but propagates recovered classicality through the accessible range. Parallel deletion preserves the recovery condition, whereas deleting a unique representative from

regular evidence destroys completeness and forces the remaining realized state to **N**. These are structural recovery theorems, not a completeness theorem for classical logic or a claim that arbitrary epistemic evidence is matroidal.

Gate 6 lifts the recovered sector compositionally. Classical atomic valuations force every belief-free propositional formula to remain in $\{\mathbf{T}, \mathbf{F}\}$, yielding strict formula-level excluded middle and a restricted formula-level explosion result. Primitive raw possibility and evidence-stable knowledge preserve classical accessible profiles, so every belief-free modal formula built from these operators remains classical under globally classical atoms. Lockean belief is the boundary: a verified two-world 1/2–1/2 profile at threshold 3/5 reopens **N** despite classical atoms. Belief is classical exactly when its two threshold decisions are complete and consistent, and a recursive admissibility predicate lifts this condition to the full legacy formula syntax.

Gate 7 derives the missing consistency half from the stronger finite-probability semantics. On a classical accessible profile, probability integrity makes positive and negative support masses complementary, so their sum is one. The model's built-in $c > 1/2$ threshold then forbids simultaneous positive and negative acceptance. Classical belief therefore reduces exactly to threshold completeness, and nonclassical belief in this sector is exactly indecision. A model-independent threshold analysis shows that $c \leq 1/2$ guarantees completeness while $c > 1/2$ guarantees consistency; the exact 1/2–1/2 tie is glut-like on the first side and gap-like on the second. Consequently no inclusive Lockean threshold makes every complementary profile threshold-regular. Under probability integrity the recursive formula-recovery contract can therefore drop the separate consistency obligation at belief nodes. Decisiveness itself remains a genuine semantic condition.

Gate 8 gives the repository's existing split CPEL translation an explicit Boolean target semantics. For every legacy formula, the positive and negative translations reproduce the two FDE support bits exactly, yielding an unconditional pointwise representation of all four values by two Boolean coordinates. The classical sector is characterized exactly by the complement equation between those coordinates: **T** and **F** form the discrete anti-diagonal of the four-valued evidence square. Under Gate 7 probability recovery, the negative translation is therefore determined by the positive one and the full recovered FDE value is reconstructible from one Boolean target bit. LP and strict-to-tolerant consequence are also represented exactly in the evaluator induced by the same 4-PEL models, and source ST derivability transports soundly. This is a semantic representation and consequence bridge, not a CPEL completeness, canonical-model, or decidability result.

The Lean 4.31 artifact distinguishes the lightweight model interface from probability-integrity-certified models and records declaration-level axiom dependencies, rejecting native evaluation in the selected manuscript proof chains. The contribution is a mechanized case study with explicit preservation and information-loss boundaries; no completeness, general continuity, or priority claim is made.

1 Introduction

Probabilistic epistemic models are usually asked two different questions. One concerns degrees: how much support does an agent assign to a proposition? The other concerns categories: does the agent believe, know, reject, or remain undecided? Threshold approaches connect the two by turning sufficiently high probability into all-or-nothing belief. Four-valued semantics adds a second axis. Positive and negative support can coexist or both fail, so the categorical result is no longer binary but belongs to

$$\{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\} = \{(1,0), (0,1), (1,1), (0,0)\},$$

following the Belnap–Dunn/FDE tradition [1, 5].

The 4-PEL project combines these ideas in a finite-accessibility evidence semantics. Positive and negative extensions are measured separately by the same local set function. A Lockean threshold is applied to each mass, producing a four-valued belief state. A separate evidence-stable knowledge operator then checks whether the complete FDE profile is invariant across the accessible range. This yields an exact factorization:

$$\mathsf{K}_i\varphi = \begin{cases} \mathsf{B}_i\varphi, & \mathsf{Stable}_i(\varphi), \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

The knowledge operator is therefore not a second probability threshold. It is a qualitative stability filter on top of threshold belief.

The lightweight interface and its stronger probability-certified subclass are distinguished in Section 2. The paper studies one-step conditionalization and, separately, a chosen affine interpolation of its support endpoints:

Research Question 1.1 (Dynamic phase geometry)**.** Which four-valued belief and knowledge endpoints can admissible conditionalization connect, and what additional phase structure follows if their signed supports are interpolated affinely?

The machine-checked answer develops in several layers. First, admissible conditionalization can both destroy and restore the stability gate. More strongly, a fixed six-world construction scheme realizes every ordered pair of four-valued values at the level of $\mathsf{K}(\mathsf{B}p)$. Thus the categorical reachability graph is complete. That global permissiveness is paired with a sharp local invariant: a belief value changes exactly when at least one of its two threshold decisions changes.

The four categorical states can consequently be treated as the vertices of a Boolean square. The number of changed threshold coordinates defines a wall count

$$d_{\text{thr}} \in \{0, 1, 2\}.$$

Zero walls means the same FDE state, one wall means an edge move, and two walls mean a diagonal move. The two diagonals are $\mathbf{T} \leftrightarrow \mathbf{F}$ and $\mathbf{N} \leftrightarrow \mathbf{B}$.

The geometric interpretation can then be reconnected to the underlying rational support masses. A changed threshold bit is exactly endpoint straddling of the Lockean boundary. For the explicit affine interpolation

$$\gamma(t) = x + t(y - x), \qquad t \in \mathbb{Q},$$

straddling produces a rational crossing parameter $t \in [0, 1]$ at which $\gamma(t) = c$. A two-wall transition therefore contains two threshold events, one for positive support and one for negative support. Each affine crossing time is unique. Their order is positive-first, simultaneous, or negative-first, and the simultaneous case is exactly passage through the point (c, c) in the support plane.

If the two crossings are not simultaneous, the rational midpoint between them occupies a categorical state adjacent to both diagonal endpoints. The order of the wall events determines the intermediate state. For example,

$$\mathbf{N} \to \mathbf{B}$$

becomes either

$$\mathbf{N} \to \mathbf{T} \to \mathbf{B}$$

or

$$\mathbf{N} \to \mathbf{F} \to \mathbf{B}.$$

The same classification holds for the other diagonal orientations. Here "time" means the interpolation parameter, not elapsed time or an inferred intermediate stage of a one-step update. Wall count counts differing endpoint bits, not crossings along an arbitrary path.

A separate verified layer constructs complete probability-integrity-certified models. Non-negative normalized rational point weights generate local measures satisfying an explicit finite-probability integrity contract. Convex interpolation preserves those distributions and makes the mass of every fixed event affine in the interpolation parameter. The local paths can then be packaged into complete strong 4-PEL models while worlds, accessibility, atomic valuation, and threshold remain fixed.

For fixed events this yields a compact structural picture; applying it to the support of a varying modal formula requires an additional bridge:

$$\boxed{\begin{array}{c}\text{probability simplex} \longrightarrow \text{strong model path}\\ \longrightarrow \text{affine fixed-event masses} \longrightarrow \text{threshold sides}\\ \longrightarrow \text{wall crossings} \longrightarrow \text{crossing order}\\ \longrightarrow \text{FDE phase sequence.}\end{array}}$$

Reliability refinement. Section 13 studies a second information-loss map: six reliability-tagged evidence cells project to four coarse cells. Projection preserves support and mixing, but not the recoverability of reliability. Fine modal stability is equivalent to coarse stability plus local fibre rigidity. These elementary semantic results are not a six-valued logical calculus; they extend the paper's investigation of which structure a coarse observation forgets.

Methodological discipline. The paper distinguishes four levels throughout. Machine-checked theorems are reported as such. Finite witness constructions are not silently generalized beyond their quantified statement. Geometric and philosophical readings are identified as interpretations. Literature and novelty claims remain deliberately modest until a systematic comparison has been completed.

Current boundary. Model-valued convex paths are now verified for weight-generated strong endpoint models on a common semantic skeleton. This does not yet imply that every legacy conditionalization witness has such a strong representation, that every intermediate point is itself generated by conditionalization of the source model, or that support events defined by arbitrary probability-sensitive modal formulas remain fixed along the path. The rational affine crossing theorem also remains distinct from an abstract intermediate-value theorem over arbitrary continuous paths. These formula-level, update-level, and topological questions are the next mathematical boundaries.

2 Four-Valued Probabilistic Epistemic Semantics

The formal development starts with a lightweight, finite-accessibility interface

$$M = (W, R, \mu, v, c),$$

where W is the world type, $R_i(w)$ is a finite accessibility list, $\mu_{i,w}$ is a rational-valued function on event lists, v assigns FDE values to atoms, and c_i satisfies

$$\frac{1}{2} < c_i \leq 1.$$

The interface requires only $\mu_{i,w}(R_i(w)) = 1$ and $\mu_{i,w}(\varnothing) = 0$, besides the threshold bounds. It does *not* require nonnegativity, additivity, extensionality, or duplicate-free lists. Consequently a general `Model` is not yet a certified probability model; "support mass" below means its rational set-function value unless integrity is explicitly assumed. The stronger `StrongProbabilityMode l` contract in Section 9 supplies ordinary finite-measure laws on accessible, duplicate-free events. Both signed masses then lie in $[0, 1]$.

The stored `worlds` list need not exhaust the type W, and the interface does not enforce accessibility closure into that list. The displayed finite witnesses do use exhaustive finite types and closed ranges. Generic theorems quantify over the literal Lean interface, not over an unstated finite-universe axiom. No reflexivity, transitivity, or Euclidean condition is generally assumed. Normalization does exclude an empty accessible range: otherwise its mass would be both zero and one (`PaperReview.accessibility_nonempty`).

2.1 Bilateral four-valued information

A semantic value is a pair of Boolean coordinates

$$x = (x^+, x^-),$$

with the four canonical values

$$\mathbf{T} = (1,0), \qquad \mathbf{F} = (0,1), \qquad \mathbf{B} = (1,1), \qquad \mathbf{N} = (0,0).$$

The positive coordinate records support for the formula and the negative coordinate records support against it. Thus $\mathbf{B}$ is a glut and $\mathbf{N}$ is a gap. Negation exchanges the two coordinates, while conjunction takes conjunction on positive support and disjunction on negative support. Independence here means that the coordinates are not logical complements; it does not assert stochastic independence of their support events. In an integrity-certified model the two events share one measure and may overlap.

This bilateral representation is crucial for the dynamic analysis. The two coordinates can cross a probabilistic threshold at different times, producing a temporal structure that would be invisible in a single Boolean belief status.

2.2 Threshold belief

For a modal formula φ, define its positive and negative accessible extensions at (i, w) by

$$E^+_{i,w}(\varphi) = \{u \in R_i(w) : [\![\varphi]\!]^+_u = 1\},$$

$$E^-_{i,w}(\varphi) = \{u \in R_i(w) : [\![\varphi]\!]^-_u = 1\}.$$

The corresponding support masses are

$$P^+_{i,w}(\varphi) = \mu_{i,w}(E^+_{i,w}(\varphi)), \qquad P^-_{i,w}(\varphi) = \mu_{i,w}(E^-_{i,w}(\varphi)).$$

Probabilistic belief thresholds the two masses independently:

$$\mathsf{B}_i\varphi = \Big([P^+_{i,w}(\varphi) \geq c_i], [P^-_{i,w}(\varphi) \geq c_i]\Big).$$

Accordingly, $\mathsf{B}_i\varphi = \mathbf{B}$ is possible when both signed support masses cross the threshold, while $\mathsf{B}_i\varphi = \mathbf{N}$ occurs when neither does.

2.3 Evidence-stable knowledge

The modal object language contains primitive probabilistic belief B, primitive evidence-stable knowledge K, and primitive raw accessibility possibility. The latter is kept distinct from the internal expression $\neg\mathsf{K}\neg\varphi$, because the two need not agree in a four-valued setting.

For the present paper, the central modal predicate is full-value stability. Let

$$\mathsf{Stable}^M_i(w, \varphi)$$

mean pairwise homogeneity on the accessible range:

$$H_i(w, \varphi) \iff \forall u, z \in R_i(w), \quad [\![\varphi]\!]_u = [\![\varphi]\!]_z.$$

Here $H_i = \mathsf{Stable}_i^M$. This does not compare the range with the current world unless that world is accessible. The primitive definition, writing $x_u = [\![\varphi]\!]_u$, is

$$(\mathsf{K}_i\varphi)^+ = H_i \wedge \forall u \in R_i(w)\ x_u^+, \qquad (\mathsf{K}_i\varphi)^- = \neg H_i \vee \exists u \in R_i(w)\ x_u^-.$$

These are classical meta-level Boolean operations on evidence coordinates. In particular, primitive K does not directly inspect μ or c. Heterogeneity is stipulated to count as negative evidence for this operator. Without frame assumptions no factivity principle at the current world is inferred; even on reflexive frames, a glut is not classical truth only.

The resulting relationship between knowledge and belief is exact.

Theorem 2.1 (Knowledge as stability-filtered belief)**.** *For every finite 4-PEL model, agent, world, and modal formula,*

$$\mathsf{K}_i\varphi = \begin{cases} \mathsf{B}_i\varphi, & \mathsf{Stable}_i(\varphi), \\ \mathbf{F}, & \neg\mathsf{Stable}_i(\varphi). \end{cases}$$

Status: Proved; trust inventory in Appendix A.

Proof sketch. The accessible range is nonempty. If homogeneous, each signed extension is either the entire range or empty. Normalization and $0 < c_i \leq 1$ make its threshold bit equal the corresponding common evidence bit. The definition of K returns those same bits. If heterogeneous, its positive bit is false and its negative bit true. No additivity assumption is needed. □

Hence an unstable accessible profile is not merely low-confidence evidence. It is a structural failure mode that absorbs every probabilistic belief value into strict $\mathbf{F}$ at the knowledge layer. Conversely, on a homogeneous profile the stability filter is transparent and $\mathsf{K}_i\varphi$ preserves the complete four-valued belief state.

For a fixed interpretation $u \mapsto x_u$ and fixed accessible range, changing only probability cannot change K. This is verified separately by `PaperReview.knowledge_ignores_probability_for_fixed_values`. Probability affects $\mathsf{K}\varphi$ only through probability-sensitive subformulas of φ; the two-case factorization is not a claim of two independent numerical inputs to primitive knowledge.

3 Admissible Conditionalization and Dynamic Knowledge Phases

The update operator conditions each local set function on the prior positive extension of an evidence formula E in the legacy language (atoms, negation, conjunction, and belief). It is not currently defined for arbitrary formulas with primitive K or raw possibility. Let

$$E_{i,w}^+ = \{u \in R_i(w) : [\![E]\!]_u^+ = 1\}.$$

For a finite set $S \subseteq W$, the raw updated local probability is

$$\mu_{i,w}^E(S) = \frac{\mu_{i,w}(S \cap E_{i,w}^+)}{\mu_{i,w}(E_{i,w}^+)}$$

whenever the denominator is nonzero. At the model level, the Lean constructor requires an explicit *ConditionalizationAdmissible* proof. Its field named `positive_mass` literally asserts *nonzero* mass at every agent/world pair, not strict positivity. Nonzero becomes positive when nonnegativity is separately available. Two further proof fields require the updated total mass to be one and its empty-event mass zero. They do not certify all strong probability laws. The event is evaluated in the prior model and held fixed for this update; this is not iterative reevaluation of its formula.

The update is intentionally conservative:

$$W^E = W, \qquad R^E = R, \qquad v^E = v, \qquad c^E = c,$$

and only μ changes. Thus any dynamic change in a formula must ultimately be mediated by a semantic dependency on probability.

3.1 A syntactic invariant sector

The modal language has a probability-free fragment containing atoms, negation, conjunction, primitive knowledge, and raw possibility, but no occurrence of the probabilistic belief constructor. Lean proves that every such formula preserves its complete FDE value under admissible conditionalization. Its accessible stability profile, outer knowledge, and raw possibility are therefore preserved as well.

This result isolates a first dynamic boundary:

$$\text{no probabilistic belief occurrence} \quad \Longrightarrow \quad \text{complete value invariant under conditionalization.}$$

The converse is false. Later sections exhibit formulas containing B whose probabilities change while their threshold status remains invariant.

3.2 Two cases in the knowledge factorization

The factorization theorem from Section 2 yields four combinations of the before/after stability bits for $\mathsf{K}_i\varphi$. Let M_0 and M_1 be two models on the same modal language. The stability states of φ determine how the belief channel is exposed at the knowledge layer:

Stable_{M_0}	Stable_{M_1}	knowledge phase
true	true	$\mathsf{K} = \mathsf{B}$ on both sides
true	false	posterior $\mathsf{K} = \mathbf{F}$
false	true	prior $\mathsf{K} = \mathbf{F}$, posterior $\mathsf{K} = \mathsf{B}$
false	false	$\mathsf{K} = \mathbf{F}$ on both sides

The classification is update-independent: it follows directly from the pointwise factorization. The conditionalization development additionally contains finite witnesses for both off-diagonal phases. One update fractures a homogeneous $\mathsf{B}p$ profile into a heterogeneous profile, giving

$$\mathsf{Stable}(\mathsf{B}p): \text{ true} \to \text{false}, \qquad \mathsf{K}(\mathsf{B}p): \ \mathbf{T} \to \mathbf{F}.$$

A second witness restores stability in the opposite direction:

$$\mathsf{Stable}(\mathsf{B}p): \text{ false} \to \text{true}, \qquad \mathsf{K}(\mathsf{B}p): \ \mathbf{F} \to \mathbf{T}.$$

Corollary 3.1 (Truth-only status can be lost and regained)**.** *There are admissible conditionalizations taking* $\mathsf{K}(\mathsf{B}p)$ *from* $\mathbf{T}$ *to* $\mathbf{F}$*, and others from* $\mathbf{F}$ *to* $\mathbf{T}$*. Thus preservation of the truth-only status is not an unconditional update law. This is not a monotonicity theorem about unspecified truth or information orders.* Status: Finite-witness; trust inventory in Appendix A.

This phase table motivates the stronger question addressed next. Instead of asking only whether stability can fracture or recover, one can ask which complete four-valued endpoints are reachable when the belief profile is kept homogeneous so that the outer knowledge filter remains transparent.

4 Complete Dynamic Epistemic Reachability

The strongest endpoint theorem uses a uniform six-world construction scheme. For arbitrary source and target values

$$s, t \in \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

consider worlds

$$\{f, a_1, a_2, a_3, a_4, r\}.$$

The distinguished focus world f carries the target value t for atom p, each of the four anchor worlds a_k carries the source value s, and the spare world r carries $\mathbf{N}$. A second atom e is $\mathbf{T}$ only at f and $\mathbf{F}$ elsewhere. Every world accesses all six worlds. The prior measure is uniform,

$$\mu(\{w\}) = \frac{1}{6},$$

and the Lockean threshold is

$$c = \frac{2}{3}.$$

Before update, the four anchors contribute total mass $4/6 = 2/3$. Coordinate by coordinate, they therefore force the threshold belief $\mathsf{B}p$ to reproduce the source value s, regardless of the single target-bearing focus world. After conditioning on e, all posterior mass is concentrated at f, so $\mathsf{B}p$ reproduces the target value t.

The local probability function is independent of the evaluation world. Consequently, the complete $\mathsf{B}p$ profile is homogeneous across the common accessible range before and after update. The outer knowledge operator therefore passes both values through unchanged.

Theorem 4.1 (Complete dynamic epistemic reachability)**.** *For every ordered pair (s, t) of complete FDE values, there is a member of the six-world family and an admissible conditionalization such that*

$$\mathsf{K}(\mathsf{B}p) : s \longrightarrow t.$$

Across that update the accessibility relation, atomic valuation, and threshold are fixed; only the local probability field changes. Status: Finite-witness; trust inventory in Appendix A.

Proof sketch. For either sign, a true source bit contributes at least $4/6 = c$, while a false source bit receives at most $1/6 < c$ from the focus. The evidence has mass $1/6$; conditioning gives focus mass one. Homogeneity of the inner belief value makes each of its outer signed support events either the whole range or empty, so $\mathsf{K}(\mathsf{B}p)$ has the asserted endpoints. The source bits equal to the threshold use the inclusive $\geq$ convention. □

Equivalently, the categorical reachability graph is complete:

source \ target	**T**	**F**	**B**	**N**
T	✓	✓	✓	✓
F	✓	✓	✓	✓
B	✓	✓	✓	✓
N	✓	✓	✓	✓

The theorem immediately gives creation and removal of both nonclassical statuses. In particular, the development contains verified instances

$$\mathbf{T} \to \mathbf{B}, \qquad \mathbf{B} \to \mathbf{N}, \qquad \mathbf{N} \to \mathbf{T}.$$

Thus neither glutty nor gappy knowledge is dynamically terminal under the present update semantics.

Quantifier discipline. The theorem does not say that one fixed concrete model realizes all sixteen arrows. The family is parameterized by the requested source and target values. For each pair, however, the pre-update and post-update models share the same R, v, and c, and the transition is generated by admissible probability-only conditionalization. The result is therefore a semantic non-prohibition theorem: no ordered pair of complete four-valued knowledge statuses is ruled out merely by the conditionalization mechanism.

The uniform and point-mass measures have the intended finite-probability reading, but the current Lean reachability statement returns the lightweight `Model`, not a `StrongProbabilityModel` certificate. Its separate strong-promotion obligation is not discharged by the generic convex-path constructor.

The reachability theorem establishes permissiveness across the witness family. The next result identifies the exact local condition under which no categorical change occurs.

5 Threshold-Side Robustness and the Boolean Square

Complete reachability shows how permissive the dynamics can be across different models in the witness family. The complementary question is local: what must change in order for one probabilistic belief value to change at all?

Let

$$p^+ = P^+_{i,w}(\varphi), \qquad p^- = P^-_{i,w}(\varphi).$$

Then the complete belief value is exactly the pair of threshold decisions

$$\mathsf{B}_i\varphi = ([p^+ \geq c_i], [p^- \geq c_i]).$$

This gives an immediate semantic boundary.

Theorem 5.1 (Threshold-side robustness). *For any two models on the same language, at a fixed agent, world, and formula,*

$$\mathsf{B}_i^{\text{after}}\varphi = \mathsf{B}_i^{\text{before}}\varphi$$

if and only if the positive threshold decision is unchanged and the negative threshold decision is unchanged. Status: Proved; trust inventory in Appendix A.

This equivalence is immediate from equality of Boolean pairs; it is a useful interface lemma, not a new probability law. In the general two-model theorem, each model uses its own threshold. Under conditionalization the threshold is fixed. The raw masses may therefore move substantially while the categorical result remains fixed. The invariant is not the precise support mass but the pair of threshold sides occupied by the two masses.

The Lean development gives a sufficient compositional certificate (not a complete characterization of invariance for all formulas), `ModalConditionalizationRobust`. At a belief node, robustness requires only that the two threshold decisions be preserved at every world. The embedded formula itself need not be dynamically invariant. Negation, conjunction, evidence-stable knowledge, and raw possibility propagate robustness compositionally. The previously verified probability-free fragment is contained in this semantic class, but the inclusion is strict: a diagonal member of the reachability family gives a formula containing B whose probability distribution changes nontrivially while the belief value stays fixed.

5.1 The threshold square

Since every FDE value is already a pair of threshold bits, the categorical state space is the Boolean square

$$\mathbf{N} = (0,0), \qquad \mathbf{T} = (1,0), \qquad \mathbf{F} = (0,1), \qquad \mathbf{B} = (1,1).$$

$$\begin{tikzcd}
(0,1) & (1,1) \\
\mathbf{F} \arrow[r, dash] \arrow[d, dash] \arrow[dr, dash, dashed] & \mathbf{B} \arrow[d, dash] \arrow[dl, dash, dashed] \\
\mathbf{N} \arrow[r, dash] & \mathbf{T} \\
(0,0) & (1,0)
\end{tikzcd}$$

The solid edges indicate undirected adjacency, not a preferred update direction or a factivity ordering. Define the endpoint threshold-wall count

$$d_{\mathrm{thr}}(a,b) = \mathbf{1}[a^+ \neq b^+] + \mathbf{1}[a^- \neq b^-].$$

Lean verifies the basic geometry:

$$0 \leq d_{\mathrm{thr}}(a,b) \leq 2,$$

$$d_{\mathrm{thr}}(a,b) = 0 \iff a = b,$$

and

$$d_{\mathrm{thr}}(a,b) = 2 \iff a^+ \neq b^+ \ \wedge \ a^- \neq b^-.$$

Thus one-wall changes are exactly edge moves and two-wall changes are exactly diagonal moves. The two diagonal pairs are

$$\mathbf{N} \leftrightarrow \mathbf{B}, \qquad \mathbf{T} \leftrightarrow \mathbf{F}.$$

Proposition 5.2 (Robustness is zero displacement)**.** *Every compositionally robust modal formula has threshold-wall count zero under its designated admissible conditionalization.* Status: Proved; trust inventory in Appendix A.

The reachability construction realizes every displacement class. Hence the dynamic picture combines two properties that might initially look opposed:

global categorical reachability is complete,

while

local categorical change is exactly coordinatewise threshold change.

This tension is the starting point for the geometric refinement in the next sections.

6 From Threshold Straddling to Literal Affine Crossings

The Boolean square records which threshold decisions differ, but it does not yet say what numerical event lies behind a changed bit. The next formal layer reconnects categorical displacement to rational support masses.

6.1 Endpoint straddling

For a rational threshold c, define

$$\mathrm{Straddle}(c; x, y)$$

to mean that exactly one endpoint lies on or above the threshold:

$$(c \leq x \wedge \neg(c \leq y)) \ \vee \ (c \leq y \wedge \neg(c \leq x)).$$

For rational numbers this is the expected endpoint pattern

$$x < c \leq y \quad \text{or} \quad y < c \leq x.$$

Theorem 6.1 (Threshold decision versus endpoint straddling)**.** *For rational* c, x, y*,*

$$[c \leq x] \neq [c \leq y] \iff \mathrm{Straddle}(c; x, y).$$

Status: Proved; trust inventory in Appendix A.

At belief level, the total wall count decomposes into the one-dimensional wall counts of the positive and negative support coordinates. Lean therefore verifies the exact three-way correspondence:

$$\begin{aligned} d_{\mathrm{thr}} = 0 &\iff \text{neither support coordinate straddles,} \\ d_{\mathrm{thr}} = 1 &\iff \text{exactly one support coordinate straddles,} \\ d_{\mathrm{thr}} = 2 &\iff \text{both support coordinates straddle.} \end{aligned}$$

In particular, every genuine categorical belief change has a numerical endpoint witness on at least one signed support axis.

6.2 Affine interpolation

Endpoint straddling still describes only two snapshots. To expose an actual crossing event, the formalization introduces the directed rational affine path

$$\gamma_{x,y}(t) = x + t(y - x), \qquad t \in \mathbb{Q}.$$

It satisfies

$$\gamma_{x,y}(0) = x, \qquad \gamma_{x,y}(1) = y.$$

For the increasing orientation $x < c \leq y$, the explicit threshold parameter is

$$t_+ = \frac{c - x}{y - x}.$$

For the decreasing orientation $y < c \leq x$, the development uses the equivalent directed parameter

$$t_- = \frac{x - c}{x - y}.$$

Lean Core rational arithmetic is sufficient to prove that the relevant parameter lies in the unit interval and hits the threshold exactly.

Theorem 6.2 (Affine unit-interval threshold crossing)**.** *If rational endpoints* x, y *straddle rational threshold* c*, then*

$$\exists t \in \mathbb{Q} : 0 \leq t \leq 1 \quad \wedge \quad \gamma_{x,y}(t) = c.$$

Status: Proved; trust inventory in Appendix A.

Proof sketch. Straddling implies distinct endpoints. Solve $x + t(y - x) = c$ for $t = (c - x)/(y - x)$; the straddling inequalities give $0 \leq t \leq 1$, accounting for the sign of the denominator. Substitution verifies the hit. A hit can be at an endpoint; the theorem does not require $0 < t < 1$. □

No abstract topology is used, and there is no Mathlib dependency.

Corollary 6.3 (Belief change has a literal affine wall hit)**.** *If admissible conditionalization changes the complete FDE value of* $\mathsf{B}_i\varphi$*, then at least one of the two chosen interpolations of its endpoint support masses*

$$t \mapsto P_t^+(\varphi), \qquad t \mapsto P_t^-(\varphi)$$

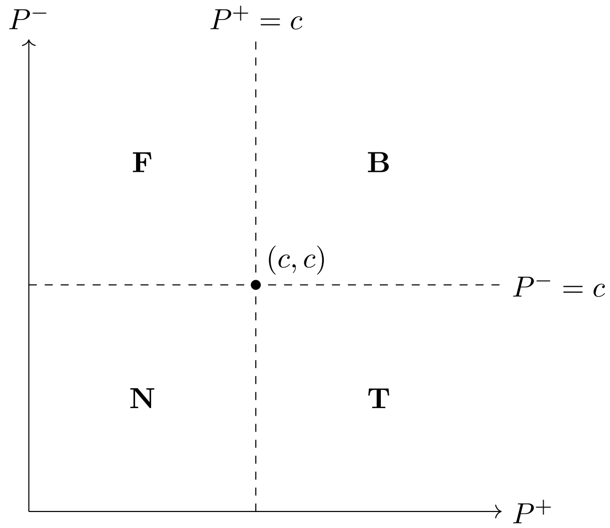
hits c_i *at some rational* $t \in [0, 1]$*. Here* $P_t^+ = (1 - t)P_0^+ + tP_1^+$*, and likewise for negative support; this notation does not yet denote evaluation in an intermediate model.* Status: Proved; trust inventory in Appendix A.

For a two-wall transition, both signed support coordinates have crossing witnesses, though they need not cross at the same parameter. This asks an order question about the chosen interpolation: which wall is hit first?

Support-plane picture. The two masses define a point

$$(P^+, P^-) \in \mathbb{Q}^2.$$

The threshold lines $P^+ = c$ and $P^- = c$ partition the plane into the four FDE regions.



The threshold square is the four-region categorical observation of the signed support plane. These four regions must not be confused with the four evidence masses (t, b, n, f) in a probability simplex. For integrity-certified models the relevant support points lie in the unit square; the algebraic lemmas themselves permit arbitrary rational coordinates.

7 Unique Crossing Times and Temporal Order

A two-wall transition gives one affine threshold hit on the positive support coordinate and one on the negative coordinate. To turn these hits into a temporal classification, the crossing parameter must be intrinsic rather than an arbitrary existential witness.

7.1 Uniqueness

Suppose $x \neq y$. The affine map

$$\gamma_{x,y}(t) = x + t(y - x)$$

is injective in its parameter. Lean proves this directly over the rationals by cancelling the nonzero factor $y - x$.

Theorem 7.1 (Unique affine threshold time)**.** *Let $x \neq y$. If*

$$\gamma_{x,y}(t) = c \qquad \textit{and} \qquad \gamma_{x,y}(s) = c,$$

then

$$t = s.$$

Consequently every straddling affine segment has a unique threshold crossing parameter in $[0, 1] \cap \mathbb{Q}$. STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

For a two-wall belief transition write

$$t_+ = \text{the unique positive-support crossing time},$$

$$t_- = \text{the unique negative-support crossing time}.$$

Both lie in the rational unit interval.

7.2 Crossing-order trichotomy

The linear order on $\mathbb{Q}$ now yields an exhaustive classification:

$$t_+ < t_-, \qquad t_+ = t_-, \qquad t_- < t_+.$$

Theorem 7.2 (Affine crossing-order trichotomy)**.** *Every two-wall conditionalized belief transition admits intrinsic positive and negative crossing times satisfying exactly one of*

$$t_+ < t_-, \qquad t_+ = t_-, \qquad t_- < t_+.$$

Status: Proved; trust inventory in Appendix A.

The three cases have immediate event readings:

$$\begin{aligned} t_+ < t_- &\ :\ \text{positive threshold wall first,} \\ t_+ = t_- &\ :\ \text{simultaneous crossing,} \\ t_- < t_+ &\ :\ \text{negative threshold wall first.} \end{aligned}$$

7.3 The simultaneous case

The equality case has a geometric characterization stronger than mere coincidence of two stored parameters.

Theorem 7.3 (Simultaneity as the wall intersection)**.** *For a two-coordinate affine crossing pair,*

$$t_+ = t_-$$

if and only if there exists one rational parameter t such that

$$P_t^+ = c \qquad \textit{and} \qquad P_t^- = c.$$

Equivalently, the support path passes through the wall-intersection point

$$(c, c).$$

Status: Proved; trust inventory in Appendix A.

Thus simultaneity requires the explicit equality of the two crossing parameters. No probability-zero or genericity theorem about this condition is claimed. If $t_+ \neq t_-$, there is a nonempty rational interval between the two wall events. The next section proves that this interval carries a forced categorical phase.

Why order matters. Categorical endpoint values alone do not specify which threshold coordinate changes first along a chosen affine segment. Yet for a diagonal transition both bits must change. The temporal order of those changes therefore contains information absent from the pair $(\text{source}, \text{target})$. Crossing-order geometry recovers exactly that lost information.

8 Intermediate FDE Phases Between Sequential Crossings

Suppose a two-wall transition is not simultaneous. The two unique crossing times then determine an open interval in which exactly one threshold coordinate has already switched to its target side.

The formal development chooses the rational midpoint

$$t_m = \frac{t_+ + t_-}{2}.$$

If $t_+ < t_-$, Lean proves

$$t_+ < t_m < t_-.$$

If $t_- < t_+$, the symmetric inequalities hold. These midpoint facts require only rational arithmetic.

8.1 Threshold side before and after a crossing

The affine support coordinate is strictly monotone whenever its endpoints are distinct. For an increasing straddling path, parameters before the unique crossing remain below threshold and parameters after it lie above threshold. For a decreasing path, the threshold sides reverse. The formal result can be stated without choosing an orientation:

Proposition 8.1 (Source side before, target side after)**.** *Let a rational affine support coordinate straddle c and hit c uniquely at time t_c. Then*

$$t < t_c \quad \Longrightarrow \quad \mathrm{side}_c(\gamma(t)) = \mathrm{side}_c(x),$$

and

$$t_c < t \quad \Longrightarrow \quad \mathrm{side}_c(\gamma(t)) = \mathrm{side}_c(y).$$

Status: Proved; trust inventory in Appendix A.

This is the key bridge from event order to a categorical intermediate state.

8.2 Generic intermediate vertices

Let

$$s = (s^+, s^-), \qquad t = (t^+, t^-)$$

be opposite vertices of the threshold square. Define

$$I_+(s,t) = (t^+, s^-)$$

for positive-first motion and

$$I_-(s,t) = (s^+, t^-)$$

for negative-first motion.

Theorem 8.2 (Positive-first midpoint phase)**.** *If both support coordinates straddle and $t_+ < t_-$, then at the rational midpoint*

$$\mathit{state}(t_m) = I_+(s,t).$$

That is, the positive coordinate has already reached the target threshold side while the negative coordinate is still on the source threshold side. Status: Proved; trust inventory in Appendix A.

Theorem 8.3 (Negative-first midpoint phase)**.** *If both support coordinates straddle and $t_- < t_+$, then*

$$\mathit{state}(t_m) = I_-(s,t).$$

Status: Proved; trust inventory in Appendix A.

For opposite endpoints, both intermediate states are adjacent to source and target:

$$d_{\mathrm{thr}}(s, I_\pm(s,t)) = 1, \qquad d_{\mathrm{thr}}(I_\pm(s,t), t) = 1.$$

Hence every sequential diagonal displacement decomposes into two one-wall phases.

Boundary values at a simultaneous hit. The threshold convention is inclusive. At the common hit (c,c) the value is always **B**, verified by `PaperReview.threshold_intersection_is_B`. Consequently "no intermediate open phase" does not mean "no intermediate value at any parameter." For example, at $c = 3/5$, the segment from $(4/5, 2/5)$ to $(2/5, 4/5)$ has **T** at zero, **B** at $1/2$, and **F** at one. The common hit is an isolated **B** between the two open phases; `PaperReview.simultaneous_T_F_has_B_hit` checks the three values. The theorem about a sequential midpoint does not include the crossing instants.

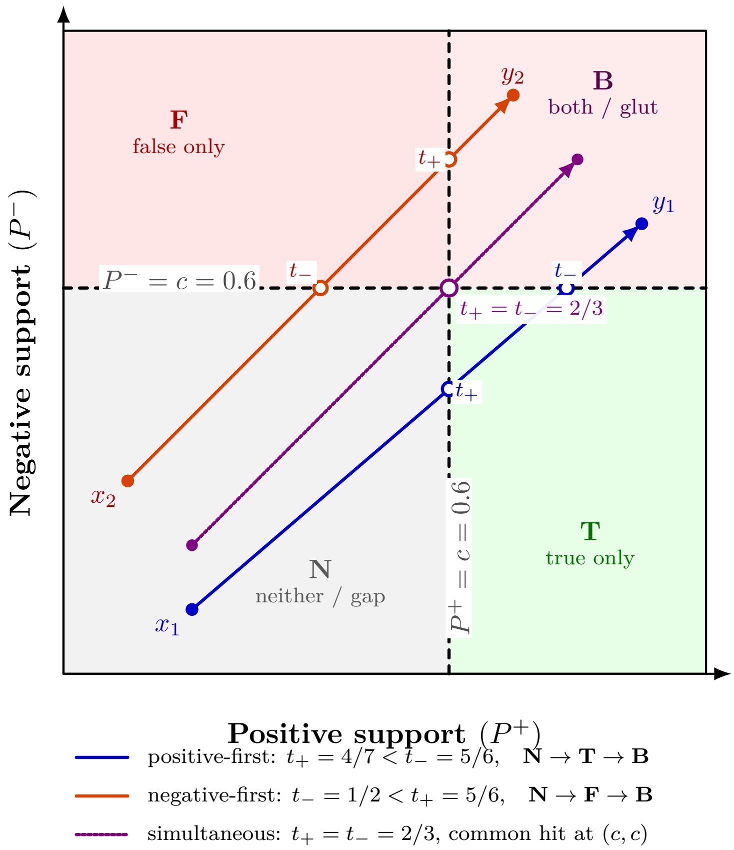


Figure 1: Affine phase geometry for three $\mathbf{N} \to \mathbf{B}$ support trajectories at threshold $c = 0.6$. Positive-first and negative-first paths necessarily occupy $\mathbf{T}$ and $\mathbf{F}$, respectively, between their unique wall-crossing times. The simultaneous path is the exceptional case $t_+ = t_-$ and passes through the wall intersection (c, c). The figure visualizes the Lean-verified crossing-order and intermediate-phase theorems; it does not by itself assert that every interpolated support point is already realized by a full admissible model.

8.3 Concrete diagonal table

The complete verified table is:

Diagonal transition	positive wall first	negative wall first
$\mathbf{N} \to \mathbf{B}$	**T**	**F**
$\mathbf{B} \to \mathbf{N}$	**F**	**T**
$\mathbf{T} \to \mathbf{F}$	**N**	**B**
$\mathbf{F} \to \mathbf{T}$	**B**	**N**

Thus, for example,

$$\mathbf{N} \to \mathbf{B}$$

can have either phase sequence

$$\mathbf{N} \to \mathbf{T} \to \mathbf{B}$$

or

$$\mathbf{N} \to \mathbf{F} \to \mathbf{B},$$

and the choice is determined exactly by whether positive or negative support crosses the Lockean wall first.

Figure 1 makes the trichotomy explicit for three rational affine trajectories with the same threshold. The two sequential cases pass through different adjacent FDE cells, whereas the simultaneous path hits the unique wall intersection and therefore has no open interval carrying an intermediate adjacent phase.

Theorem 8.4 (Two-wall intermediate-phase classification)**.** *Every two-wall conditionalized belief transition admits an affine crossing pair such that either*

(i) the two crossing times are equal and the support path hits (c, c), or

(ii) the positive crossing is earlier and the midpoint has the positive-first intermediate FDE state, or

(iii) the negative crossing is earlier and the midpoint has the negative-first intermediate FDE state.

Status: Proved; trust inventory in Appendix A.

Scope of this theorem. The theorem in this section classifies the explicit affine interpolation of the two support masses. Section 9 subsequently verifies a genuine path of complete strong 4-PEL models for weight-generated endpoints and proves that every fixed event mass follows exactly such an affine trajectory. This does not realize arbitrary legacy support endpoints as a strong model path. For weight-generated endpoints the further gap is the formula-level identification of the positive and negative support events with fixed events along the model path. For arbitrary probability-sensitive modal formulas those event memberships may themselves change with the path parameter.

9 From Probability Simplexes to Strong Model Paths

The affine crossing results of the previous sections are formulated at the level of signed support masses. To interpret those trajectories as genuine probabilistic model trajectories, the probability layer must first satisfy more than the lightweight normalization contract built into the original `M odel` structure. This section records the stronger layer and the verified lift from rational weight vectors to complete 4-PEL models.

9.1 Finite probability integrity

The compatibility predicate `FiniteProbabilityIntegrity` strengthens one local measure by requiring duplicate-free support, nonnegative event masses, extensionality with respect to event membership, monotonicity, disjoint finite additivity, empty-event mass zero, and total support mass one. The wrapper `StrongProbabilityModel` equips an ordinary 4-PEL model with this stronger certification.

The stronger contract is deliberately additive rather than a replacement of the legacy model interface. Earlier finite witnesses therefore remain available, while later geometric results can state explicitly when genuine finite-probability behavior is required.

Proposition 9.1 (Weight-generated finite probability)**.** *Let R be a duplicate-free finite support and let $q : R \to \mathbb{Q}$ be nonnegative with total mass one. Define the mass of a finite event S by summing the weights of the points of R that belong to S. Then the resulting local measure satisfies* `FiniteProbabilityIntegrity`*.* Status: Proved; trust inventory in Appendix A.

The implementation iterates over the fixed support rather than over the event list. Consequently event order and duplicate presentation do not alter the generated mass.

9.2 Convexity of the finite probability simplex

For two valid weight distributions q_0, q_1 on the same support and rational $t \in [0, 1]$, define

$$q_t(x) = (1 - t)q_0(x) + tq_1(x).$$

Lean verifies that q_t is again a valid finite weight distribution. Hence the full rational line segment between two valid endpoints remains in the finite probability simplex.

More strongly, every fixed event mass is affine in the same parameter:

$$\mu_t(S) = (1 - t)\mu_0(S) + t\mu_1(S).$$

Theorem 9.2 (Affine event mass on the probability simplex)**.** *For every fixed event S, the weight-generated mass along the convex distribution path satisfies*

$$\mu_{q_t}(S) = (1 - t)\mu_{q_0}(S) + t\mu_{q_1}(S).$$

Status: Proved; trust inventory in Appendix A.

This identity is the formal bridge between the probability simplex and the affine scalar paths used in the threshold-crossing theorems.

9.3 A model-valued rational path

The probability path can be lifted to the complete model structure. Fix a world list W, an accessibility field R, an atomic valuation v, and Lockean thresholds c. Let q_0 and q_1 provide valid local weight distributions at every agent/world pair. For every rational $t \in [0, 1]$, define

$$M_t = (W, R, \mu_t, v, c),$$

where each local measure is generated from the interpolated weights q_t.

Theorem 9.3 (Convex strong model path)**.** *Every rational $t \in [0, 1]$ determines a `StrongProbabilityModel` M_t. Along the entire path,*

$$W_t = W, \qquad R_t = R, \qquad v_t = v, \qquad c_t = c,$$

and every fixed local event has affine probability mass. Status: Proved; trust inventory in Appendix A.

Thus the geometric probability path is realized by complete probabilistically certified 4-PEL models, rather than merely by pairs of numbers in a support plane.

Exact scope of the realization. Two distinctions remain essential. First, this theorem concerns weight-generated strong endpoint models on a common semantic skeleton. It does not retroactively prove that every legacy witness in the repository satisfies the stronger probability contract. Second, a convex model path is not the same thing as an update-generated path: the theorem does not claim that every intermediate M_t arises by conditionalizing the source model on some evidence event.

There is also a formula-level boundary. The affine theorem applies directly to a *fixed event S*. For an arbitrary modal formula containing probabilistic belief, the set of worlds that positively or negatively support the formula can itself vary with t. In that case the event whose mass is measured is no longer fixed, so affine event mass alone does not imply affine `modalPositiveBeliefMass` or `modalNegativeBeliefMass`.

The next formal step is therefore narrower and sharper: identify a path-invariant formula fragment, beginning with atomic propositions and probability-free formulas, and lift the verified threshold-crossing and intermediate-phase geometry to those genuine model paths.

10 Structural Interpretation: From Reachability to Phase Kinematics

Status: structural interpretation.

The formal results support a layered interpretation of four-valued epistemic change. At the support-pair level, a belief formula is represented by two rational masses; the reliability refinement in Section 13 retains further information. Thresholding projects this point into one of four regions. The resulting FDE value forgets the precise probabilities and remembers only which side of the two Lockean walls each coordinate occupies.

This can be written as a coarse-graining map

$$q_c : \mathbb{Q}^2 \longrightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

$$q_c(p^+, p^-) = ([p^+ \geq c], [p^- \geq c]).$$

The two threshold lines partition the support plane into four cells. The Boolean threshold square is the adjacency graph of these cells. In this sense the four-valued state space is not an arbitrary collection of labels. It is the quotient of a signed evidence space by two threshold predicates.

10.1 Global freedom, local constraint

Two verified results together give the central dynamic contrast:

$$\text{every ordered categorical transition is realizable,}$$

while

$$\text{every categorical change is mediated by a threshold-side change.}$$

Global reachability is therefore maximal, but local motion is not structureless. A transition leaves a trace in the signed support coordinates. Zero-wall movement is robust, one-wall movement crosses one categorical boundary, and two-wall movement crosses both.

The affine results sharpen this further. A diagonal transition does not merely differ in both bits. Along the chosen support interpolation it contains two unique wall events. Those events may coincide, or they may be temporally ordered. If they are ordered, a forced adjacent phase lies between them. The endpoint displacement therefore hides a short event history.

It is useful to call this structure *epistemic phase kinematics*. The term is not an additional theorem. It names the combination of four verified layers:

$$\text{cell membership,} \quad \text{wall count,} \quad \text{crossing time,} \quad \text{intermediate phase sequence.}$$

10.2 The exceptional role of simultaneity

For a diagonal move, sequential crossings force an intermediate adjacent phase. Simultaneity removes the nonempty open interval between the two wall hits, although an isolated boundary value can still occur. Geometrically, simultaneity requires passage through (c, c). This is an alignment condition, not a verified measure-theoretic claim about generic trajectories.

This distinction may become important in later model-level dynamics. If admissible model paths can realize the affine support interpolation, sequential diagonal changes would carry genuine intermediate epistemic states, while simultaneous changes would behave as boundary-coincident transitions.

10.3 Connection to structural transport

A broader working hypothesis in the 4-PEL project treats epistemic paradoxes as failures of structural transport. The present dynamic results fit that program in a precise but limited way. Thresholding is a projection from rich signed probabilistic information to a coarse four-valued state. Endpoint states alone do not preserve crossing order. Recovering the order requires returning to the richer support trajectory.

The paper does not claim that all epistemic paradoxes reduce to threshold crossing. Rather, this case study supplies one clean instance of a recurring pattern:

$$\text{rich structure} \longrightarrow \text{coarse projection} \longrightarrow \text{apparent categorical jump},$$

where restoring the discarded structure reveals the mechanism hidden by the projection.

The formal contribution of the present paper ends before this philosophical generalization. A general structural-transport theorem connecting threshold dynamics to the project's other paradox families remains open.

11 Lean Mechanization and Verification Status

The artifact uses Lean 4.31.0 without Mathlib. The present extension is developed on `research/cpel-classical-collapse-gate8`, stacked on the Lockean threshold phase-transition branch and the preceding classical-recovery and matroid gates; the root module `PEL4.lean` imports the relevant development. Appendix A provides the selective claim-to-declaration map for the established manuscript core. The matroid, recovery, threshold, and CPEL programs additionally carry separate declaration-level audits. Audit selections are not a certification of every declaration in the repository.

11.1 Verification contract

A successful build establishes that declarations typecheck, not that all their assumptions are proved. The manuscript inventory in `PEL4/PaperAxiomAudit.lean` and the separate MPFG inventory are checked by CI against a strict allow-list. Only `propext`, `Classical.choice`, and `Quot.sound`, or subsets thereof, may occur in selected manuscript chains; `sorryAx`, project-specific axioms, and native-decision axioms are rejected.

The matroid gates sharpen this inventory declaration by declaration. The Gate-1 channel closure and Gate-3 circuit package are axiom-free. The Gate-2 topological bridge inherits standard classical dependencies from the existing topology layer. For Gate 4, the explicit-ground deletion and contraction constructors, contraction loop characterization, finite channel matroid, unique-channel deletion profile, and nonloop-capacity theorems are axiom-free. The packaged theorem asserting FDE state preservation under deletion of a parallel representative currently uses only the same allowed standard Lean axioms; no stronger trust claim is made for it.

Gate 5 adds `PEL4/ClassicalRecovery.lean` and `PEL4/ClassicalRecoveryStructural.lean`, with a dedicated `ClassicalRecoveryAxiomAudit.lean`. The selected inventory covers the geometry–value recovery, the recovered propositional slice, separated LEM/EFQ conditions, modal persistence, deletion robustness/fragility, and the nonloop capacity comparison before and after contraction.

Gate 6 adds `PEL4/FormulaClassicalRecovery.lean`, `PEL4/ModalFormulaClassicalRecovery.lean`, and `PEL4/BeliefClassicalRecovery.lean`, with the selected declarations collected in `FormulaClassicalRecoveryAxiomAudit.lean` and the manuscript inventory. The audited claims include propositional structural induction, formula-level LEM and restricted LP explosion, the finite classical-atoms/nonclassical-belief witness, classical preservation by raw possibility and evidence-stable knowledge, the belief-free modal induction, the exact threshold-regularity characterization of classical belief output, and the recursive full-formula recovery contract.

Gate 7 adds `PEL4/LockeanThresholdPhaseTransition.lean`, `PEL4/ThresholdPhaseGeometry.lean`, and `PEL4/ProbabilityIntegrityFormulaRecovery.lean`, each with a corresponding axiom audit. The selected claims include complementary support masses for classical profiles, supermajority-derived threshold consistency, the two threshold regimes, the exact 50/50 obstruction, and the reduced recursive recovery contract under probability integrity.

Gate 8 adds `PEL4/CPELClassicalCollapse.lean`, `PEL4/CPELConsequenceBridge.lean`, and their corresponding axiom audits. The target CPEL syntax is given an explicit Boolean evaluator over the existing finite model. The audited claims cover exact positive/negative translation semantics, unconditional reconstruction of the FDE pair from the two translated Boolean channels, the classical complement-locus characterization, the Gate 7 collapse/reconstruction theorem, and exact LP/ST consequence equivalences for the induced target evaluator. Existing ST derivability is transported only through the source soundness theorem. These declarations use only the standard allow-listed Lean axioms. No target proof system, arbitrary target-model semantics, canonical model, completeness theorem, or decidability theorem is inferred from this bridge.

The review found that some general headlines inherited native-evaluation assumptions through elementary helpers. In eight source modules, the relevant finite computations were changed from `native_decide` to `decide+kernel`, without changing statements or semantic definitions. Legacy declarations outside the audited chains may still use native evaluation or project axioms; no repository-wide axiom-freedom claim is made.

11.2 Status discipline

The manuscript follows the repository verification policy:

(1) **PROVED**: the stated theorem typechecks and its audited dependency chain contains no project-specific unproved assumption. Explicit hypotheses and model proof fields remain assumptions of the theorem's quantified statement.

(2) **FINITE-WITNESS**: a finite construction or counterexample is verified. Finite domain coverage is not a theorem about arbitrary model sizes.

(3) **EXPERIMENTAL / INTERPRETIVE**: terminology, philosophical readings, and prospective extensions without a corresponding established theorem.

(4) **AXIOMATIC-PROTOTYPE**: a substantive property is postulated. No manuscript headline is promoted from this category merely because it compiles.

The proof sketches explain the main arguments for readers; Lean verifies their formal counterparts. The geometry uses rational algebra and discrete support coordinates, not an unproved topological or vector-space structure. Numerical figure coordinates illustrate established statements and are not additional model-realization certificates.

Reproducibility. From a clean clone, or after preserving local work:

```
git fetch origin
git switch research/cpel-classical-collapse-gate8
git pull --ff-only
lake build
lake env lean PEL4/CPELClassicalCollapseAxiomAudit.lean
lake env lean PEL4/CPELConsequenceBridgeAxiomAudit.lean
python3 scripts/check_mpfg_axioms.py
python3 scripts/check_paper_axioms.py --report paper-axioms.json
python3 -m unittest discover -s scripts -p 'test_*.py'
```

```
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The mutable branch is for ongoing development; use a recorded commit for an exact reproduction. Lean checking and manuscript rendering remain separate checks. The committed PDF is the rendered working artifact, not evidence of formal correctness by itself.

12 Related Work, Limits, and Open Problems

The semantic kernel belongs to the Belnap–Dunn/FDE tradition of tracking support for truth and falsity independently [1, 5]. Four-valued modal logics with bilateral relational semantics have been studied in several forms [8, 11]. The present project adds a finite probabilistic threshold layer and a primitive evidence-stable knowledge operator whose behavior is analyzed mechanically rather than assumed to coincide with a standard modal box.

The threshold connection between all-or-nothing belief and degrees of belief belongs to the broader Lockean tradition. Leitgeb's stability theory is a particularly important point of comparison because it places stability conditions at the interface between probability and categorical belief [7]. The 4-PEL notion of stability is different in type: it concerns invariance of the complete FDE value across an accessible profile and acts as a filter from threshold belief to knowledge. A careful conceptual comparison is therefore needed before stronger claims about similarity or difference are made.

The update perspective is adjacent to dynamic epistemic logic, where informational actions transform epistemic models and knowledge states [12]. The conditionalization operator studied here is narrower: it preserves worlds, accessibility, valuation, and threshold and changes only local probabilities. Conversely, the four-valued signed-support geometry developed here is not part of the standard DEL vocabulary. The manuscript reports this construction without claiming that it is novel relative to all probabilistic or nonclassical dynamic epistemic systems.

Direct probability predecessors. Klein, Majer, and Rafiee Rad [6] already develop probabilities over Belnap–Dunn logic, including axiomatization and Bayesian and Jeffrey-style updating. Thus neither probabilistic gaps/gluts nor their updating is introduced by the present paper. Bílková et al. [2] explicitly compare signed probabilities with the four-region account and give two-layered logical systems. Their languages and representation results are important comparators, not theorems inherited by our recursively nested threshold language. No translation from their full logics to `ModalFormula` is proved here.

Reliability and status modalities. Coniglio and Rodrigues [4] provide six-valued logics with classicality propagation; their six-scenario motivation predates the 2026 probability preprint of Borja Macias, Coniglio, and Hernández-Tello [3]. The latter's abstract announces probability and Jeffrey-update results over that logic. MPFG does not reproduce these results: its tags are given data, it has no classicality connective or truth tables, and it has not yet defined reliability-sensitive updates. Likewise, Petrukhin [10] gives four-valued essence/accident modalities and S5-based proof systems. Our ordinary meta-level status predicates are not those object-language operators or their completeness theorems.

This is a targeted positioning check, not an exhaustive novelty search. The comparisons use the published descriptions and, for the two-layered and six-valued works, the accessible preprint text. The 2026 probability work is used at abstract level only; no detailed equivalence or priority judgment depends on an unavailable full-text comparison.

Novelty boundary. No strong novelty claim is made in this version. The individual mathematical ingredients are elementary: Boolean pairs, threshold comparisons, affine interpolation, finite convex probability, linear order, and midpoint arguments. The research contribution currently established internally to the project is the machine-checked composition of these ingredients into one dependency chain connecting admissible probabilistic update to four-valued categorical reachability, robustness, crossing order, intermediate phase structure, and a model-valued rational probability path. A systematic literature audit is publication-critical.

12.1 Verified model-path layer and remaining realization questions

The earlier support-to-model existence problem now has a positive answer for a precise class. Given two weight-generated strong endpoint models on the same semantic skeleton, rational convex interpolation of their local weights produces a `StrongProbabilityModel` for every $t \in [0,1] \cap \mathbb{Q}$. Worlds, accessibility, valuation, and threshold remain fixed, and the probability of every fixed event varies affinely.

Two stronger realization questions remain open.

Research Question 12.1 (Formula-level support realization)**.** For which modal formulas φ are the positive and negative support events invariant along a convex strong model path, so that the verified affine fixed-event theorem applies directly to $P_t^+(\varphi)$ and $P_t^-(\varphi)$?

Atomic propositions are the natural first case because their valuation is fixed by construction. A broader probability-free fragment is plausible but requires an explicit theorem connecting path-invariance of `evalModal` to fixed support-event membership.

Research Question 12.2 (Update-generated path)**.** When can the intermediate models of a convex strong model path themselves be obtained by admissible conditionalization of the source model, possibly using a parameterized family of evidence events?

A model-valued path and an update-generated path are different notions. The convex-model-path module verifies the former and deliberately does not infer the latter. On a fixed finite universe, event-only conditioning of one fixed prior yields only finitely many posteriors (one per positive-mass event). A nonconstant affine weight segment, in contrast, has infinitely many rational points. Hence a general identification cannot hold without changing the update mechanism. This elementary obstruction is a mathematical observation here, not an additional Lean theorem. Soft evidence or likelihood updates would require a separately specified semantics.

12.2 Open problem 2: arbitrary continuous paths

The affine crossing theorem is constructive and sufficient for the present phase result, but it is not a general topological theorem. A stronger layer would define a continuous support path

$$\gamma : [0,1] \to \mathbb{R}^2$$

and prove that endpoint straddling forces intersection with the appropriate threshold wall. Standard intermediate-value reasoning would then recover a crossing theorem independent of linear interpolation.

For two-wall transitions, arbitrary paths introduce genuinely new geometry. A path may meet the two walls multiple times, touch a wall without changing side, or revisit a categorical cell. The current uniqueness and three-case temporal order are specific to nonconstant affine coordinates. A general theory would therefore need a richer notion of crossing event than the present unique hit parameter.

12.3 Open problem 3: algebraic stability

The knowledge filter is presently defined semantically by full-value homogeneity over an accessible range. An open problem is to characterize this stability intrinsically in bilattice or algebraic terms and to determine which logical operators preserve, reflect, or destroy it. Such a characterization could connect the dynamic phase table to existing algebraic semantics for four-valued modal systems.

12.4 Open problem 4: higher-dimensional phase complexes

A single belief formula contributes two threshold bits. For n tracked formulas, the naive categorical state space has $2n$ threshold coordinates and therefore resembles a Boolean hypercube. This suggests a higher-dimensional generalization of wall count, crossing order, and phase sequence. Dependencies between formulas, however, can identify or forbid regions of that cube. Determining the realizable cell complex is a natural extension of the present two-dimensional result.

12.5 Open problem 5: structural transport and paradox families

The project contains independent formal diagnoses of Lottery, Preface, Sorites, Knower, Surprise Examination, Moore/Fitch, and dynamic threshold phenomena. The working phrase "paradoxes as failures of structural transport" is currently interpretive. A genuine general theorem would require an abstract account of transformations, observations, preserved predicates, and counterexamples that subsumes several of these families without erasing their differences.

13 Modal-probabilistic refinement and information loss

This extension is a semantic refinement of evidence, not a replacement of the four-valued algebra. Its formal declarations are in `PEL4/ModalProbability/Refinement.lean` and `PEL4/ModalProbability/StatusTransport.lean`. Here "conservative" means preservation of the specified observations under projection, not proof-theoretic conservativity of a new calculus. The gate's validation status is recorded in `docs/MODAL_PROBABILISTIC_FINE_GRAINING.md`.

Definition 13.1 (Reliability-tagged evidence profile)**.** A fine profile is a nonnegative rational vector $q = (t, b, n, f, t_r, f_r)$ of total mass one. Its coarse projection is

$$\pi(q) = (t + t_r, b, n, f + f_r).$$

The reliability tags are supplied semantic data, not proofs of empirical truth. Untagged does not mean unreliable. The six cells select a subset of the earlier unrestricted `EvidenceStatus`, which also permits other tag combinations; they do not classify that whole type. The untagged lift of (t, b, n, f) is $(t, b, n, f, 0, 0)$.

Proposition 13.2 (Affine projection with a section)**.** *The untagged lift is a right inverse of π. Signed support is preserved: $P^+(q) = t + t_r + b$ and $P^-(q) = f + f_r + b$. For $0 \leq a \leq 1$, both normalized rational simplexes are closed under mixing, and the formal projection identity is*

$$\pi((1-a)q + ar) = (1-a)\pi(q) + a\pi(r).$$

Thus raw threshold belief commutes with forgetting reliability. This does not assert that thresholding is affine; the raw belief compatibility follows from the chosen definitions. Moreover, no function of the coarse profile recovers reliable mass $t_r + f_r$ in general: pure untagged truth and pure reliable truth have the same projection, but reliable masses zero and one. Status: Proved; trust inventory in Appendix A.

Proof sketch. Summing the two truth-tag coordinates and the two falsity-tag coordinates commutes with coordinatewise convex mixing by distributivity. The untagged lift sets both tag masses to zero. If reliability were a function of the coarse profile, the two pure-truth profiles would force its value to be both zero and one. □

Definition 13.3 (Status stability and local fibre rigidity)**.** For any relation R, observation v, and projection π, put

$$\begin{aligned}\mathrm{Stable}_{\mathrm{cur}}(v,w) &\iff \forall u\,(Rwu \Rightarrow v(u)=v(w)),\\ \mathrm{Rigid}(v,\pi,w) &\iff \forall u\,(Rwu \Rightarrow (\pi(v(u))=\pi(v(w)) \Rightarrow v(u)=v(w))).\end{aligned}$$

The subscript distinguishes current-world stability from the accessible-range homogeneity H used by primitive K. On reflexive frames they agree for the same observation, but no such identification is used on arbitrary frames.

Theorem 13.4 (Local stability decomposition)**.** *For any observation and projection, with no frame assumptions,*

$$\mathrm{Stable}_{\mathrm{cur}}(v,w) \iff \mathrm{Stable}_{\mathrm{cur}}(\pi\circ v,w) \wedge \mathrm{Rigid}(v,\pi,w).$$

Status: Proved; trust inventory in Appendix A.

Proof. Fine stability implies both coarse stability (apply π) and rigidity. Conversely, coarse stability puts every accessible observation in the current coarse fibre, and rigidity then identifies its fine value with the current one. □

This is an elementary local decomposition, not a test that guarantees rigidity independently. It does not assert that π is globally injective. Two universally accessible worlds carrying untagged and reliable truth witness coarse stability with fine instability, even on a reflexive, transitive, Euclidean frame. A second witness distinguishes mass stability from categorical stability: $(0,1,0,0)$ and $(0,3/4,1/4,0)$ both yield $\mathbf{B}$ at threshold $3/5$, although their glut masses differ.

For a status s, the auxiliary predicates are

$$\mathrm{Ess}_s(w) \iff (v(w)=s \Rightarrow \forall u\,(Rwu \Rightarrow v(u)=s)),$$

$$\mathrm{Acc}_s(w) \iff v(w)=s \wedge \exists u\,(Rwu \wedge v(u)\neq s).$$

Thus Ess_s may hold vacuously when the current value is not s. Its accidental counterpart requires an actual different accessible status. These are classical meta-level predicates, not additional FDE connectives. Their relationship to the existing executable stability test and K is stated separately: applying the unchanged knowledge interpretation to raw fine belief or to its coarse projection yields the same value. No identification of K with ordinary S5 necessity is made.

Finally, the existing finite-step-chain theorem transports threshold status when each edge preserves it. Cell refinement is distinct from population Finite Fine-Grainedness, and connectivity alone does not imply status preservation. Normalized reliability-sensitive updates, probability-one versus necessity under full-support assumptions, and any representation of proof-theoretic fractional values remain subsequent research targets.

14 Matroidal evidence closure

The four-valued kernel records whether positive and negative evidence are present, but it does not by itself distinguish evidential novelty from coarse repetition. This section adds a structural layer for that distinction. The claim is deliberately restricted: arbitrary bodies of epistemic evidence are *not* assumed to form matroids. Instead, the bilateral 4-PEL signature canonically induces a two-channel closure structure. The formal declarations are in `PEL4/MatroidEvidence.lean`, with a separate declaration-level trust inventory in `PEL4/MatroidEvidenceAxiomAudit.lean`.

Definition 14.1 (Two-channel evidence closure)**.** Let E be a type of evidence atoms and let $\chi : E \to \{+, -\}$ assign each atom its coarse 4-PEL polarity. For $A \subseteq E$, define

$$\mathrm{cl}_\chi(A) = \{e \in E : \exists a \in A\ \chi(a) = \chi(e)\}.$$

Thus, after coarse projection, one item of positive evidence generates the whole positive parallel class, and likewise for negative evidence. No claim is made that the underlying fine-grained sources are interchangeable.

Theorem 14.2 (4-PEL channel closure)**.** *The operator* cl_χ *is extensive, monotone, idempotent, and satisfies the closure exchange law*

$$x \in \mathrm{cl}_\chi(A \cup \{y\}), \quad x \notin \mathrm{cl}_\chi(A) \quad \Longrightarrow \quad y \in \mathrm{cl}_\chi(A \cup \{x\}).$$

Hence the coarse positive/negative partition carries a matroid closure presentation. In particular,

$$y \in \mathrm{cl}_\chi(\{x\}) \quad \Longleftrightarrow \quad \chi(x) = \chi(y).$$

Status: Lean-proved; separate declaration-level audit in `PEL4/MatroidEvidenceAxiomAudit.lean`.

Proof sketch. Extensivity and monotonicity follow immediately from retaining a witness of the same polarity. Idempotence collapses a two-step same-polarity witness to a one-step witness. For exchange, if x enters the closure only after inserting y, then the witnessing atom cannot already belong to A; consequently x and y have the same polarity, so inserting x places y in the closure. This is the two-class partition instance of the standard closure view of matroids [13, 9]. □

For a body of evidence A, write $\mathrm{Supp}_+(A)$ and $\mathrm{Supp}_-(A)$ for the existence of positive and negative atoms respectively. These two predicates recover exactly the bilateral Boolean observations used by the FDE kernel.

Proposition 14.3 (Coarse-state conservativity)**.** *For each polarity* $\sigma \in \{+, -\}$*,*

$$\mathrm{Supp}_\sigma(\mathrm{cl}_\chi(A)) \quad \Longleftrightarrow \quad \mathrm{Supp}_\sigma(A).$$

Consequently, if A realizes an FDE value by matching its positive and negative bits to these two support predicates, then $\mathrm{cl}_\chi(A)$ *realizes exactly the same value, and conversely. Matroidal closure removes coarse redundancy without changing the four-valued observation.* Status: Lean-proved; separate declaration-level audit in `PEL4/MatroidEvidenceAxiomAudit.lean`.

Proposition 14.4 (Channel-rank profile)**.** *Define the coarse channel rank of v as the number of present bilateral support bits. Then*

$$r(\mathbf{N}) = 0, \qquad r(\mathbf{T}) = r(\mathbf{F}) = 1, \qquad r(\mathbf{B}) = 2.$$

Status: Lean-proved by reduction.

The rank profile changes the interpretation of contradiction in a useful way. At this coarse structural level, $\mathbf{B}$ is not deficient in evidence: it has maximal channel rank because both independent directions are represented. Paraconsistency is therefore compatible with high evidential breadth. Rank alone is not a truth value, however. Since $\mathbf{T}$ and $\mathbf{F}$ both have rank one, polarity orientation remains essential. The rank statistic measures how many coarse evidence directions are represented, not which direction they support and not how reliable or numerous their sources are.

Scope boundary. Same-polarity atoms are parallel only after projection to the two bits observed by the FDE kernel. Two independent laboratories, two proofs, or two sensors may remain genuinely distinct at a finer epistemic level even when both contribute positive support. The present theorem therefore establishes a canonical *coarse* matroid induced by 4-PEL; it does not establish the general thesis that evidence, justification, or information always has matroid structure. Reliability tags, source identity, probabilistic weight, and inferential origin are intentionally outside this first gate.

14.1 Gate 2: the topological exchange boundary

The repository also contains a topological evidence semantics in which an algebraic interior operator induces a closure cl. That closure is extensive, monotone, idempotent, and preserves binary unions. The last fact makes the interaction with matroid exchange unusually rigid. Let

$$x \preceq_{\mathrm{cl}} y \quad \Longleftrightarrow \quad x \in \mathrm{cl}(\{y\}).$$

For a genuine point-set topology this is the specialization preorder (up to the usual orientation convention). Symmetry of this preorder is the standard R_0 separation condition [14].

Theorem 14.5 (Topological exchange characterization)**.** *For the topological closure generated by an* `InteriorSemantics`*, the following are equivalent:*

(i) the closure satisfies the Mac Lane–Steinitz exchange condition;

(ii) singleton specialization is symmetric:

$$x \in \mathrm{cl}(\{y\}) \quad \Longrightarrow \quad y \in \mathrm{cl}(\{x\}).$$

Thus, for ordinary topological spaces, the repository-level matroid exchange boundary is exactly the R_0 boundary. Status: Lean-proved in `PEL4/MatroidTopologicalBridge.lean`; declaration-level audit included.

Proof sketch. Since topological closure preserves binary unions,

$$\mathrm{cl}(A \cup \{y\}) = \mathrm{cl}(A) \cup \mathrm{cl}(\{y\}).$$

If x enters the closure only after adjoining y, then $x \in \mathrm{cl}(\{y\})$; singleton symmetry gives $y \in \mathrm{cl}(\{x\})$, hence $y \in \mathrm{cl}(A \cup \{x\})$. Conversely, exchange over the empty base sends $x \in \mathrm{cl}(\{y\})$ to $y \in \mathrm{cl}(\{x\})$, because the topological closure of the empty set is empty. □

This result distinguishes two forms of "being generated by nearby evidence." Topological closure may encode directional approximation: one point can lie in the closure of another without reciprocal dependence. Matroid exchange forbids that asymmetry at the singleton level. In a genuine T_0 topology, adding the exchange condition therefore pushes the singleton specialization relation to equality, i.e. to the T_1 boundary.

Proposition 14.6 (Sierpiński obstruction)**.** *The two-point Sierpiński topology already used as a finite witness in the 4-PEL topological semantics does not satisfy matroid exchange. At its distinguished points,*

$$\mathsf{focus} \in \mathrm{cl}(\{\mathsf{neighbour}\}), \qquad \mathsf{neighbour} \notin \mathrm{cl}(\{\mathsf{focus}\}).$$

Hence singleton specialization is asymmetric, so the topological closure fails exchange. Status: Lean-proved in `PEL4/MatroidTopologicalSierpinski.lean`.

The obstruction is internal to the existing 4-PEL topological research line; it is not an unrelated topology introduced solely to refute exchange. The same Sierpiński witness previously exposed directional boundary behaviour for $\Box$ and $\Diamond$, and Gate 2 now identifies that directionality as the precise reason the closure is not matroidal.

Proposition 14.7 (Channel topology realization)**.** *The coarse Gate-1 positive/negative channel matroid is itself topological. Define*

$$\mathrm{Int}_\chi(A) = \{e : \forall a\, (\chi(a) = \chi(e) \Rightarrow a \in A)\}.$$

This is the partition topology generated by the polarity classes, and its dual closure satisfies

$$\mathrm{cl}_{\mathrm{Int}_\chi}(A) = \mathrm{cl}_\chi(A).$$

Its singleton specialization relation is symmetric and therefore satisfies exchange. Status: Lean-proved in `PEL4/MatroidTopologicalBridge.lean`.

Gate 2 therefore locates the original channel matroid inside the intersection of the two structural programs: it is simultaneously a partition-topological closure and a matroidal closure. The Sierpiński semantics lies outside that intersection. This gives the 4-PEL evidence layer a first nontrivial structural boundary rather than merely two parallel mathematical descriptions.

Formal scope. The repository's `MatroidClosure` is intentionally a lightweight closure-axiom presentation and currently does not add a finite-character axiom. The characterization above is therefore a theorem about exactly that formal notion. On the finite carriers used by the explicit 4-PEL witnesses, finite character is automatic; infinite-carrier finitarity remains a separate extension.

Next structural questions. The next natural gate is combinatorial rather than topological: formalize matroid circuits as minimal dependent evidence sets and determine what a minimal evidential redundancy means in 4-PEL. After that, deletion and contraction can be tested as formal models of removing an evidence source and quotienting by accepted background evidence. A further question is whether the six-cell reliability refinement preserves the R_0-like exchange boundary under coarse projection.

15 Circuits as minimal evidential redundancy

Gate 2 identified the exact topological boundary at which the coarse 4-PEL closure also satisfies matroid exchange. Gate 3 turns from closure geometry to combinatorial dependence. In matroid theory a circuit is a minimal dependent set [13, 9]. For the two-channel 4-PEL projection this notion admits a particularly sharp interpretation: a circuit is a smallest body of evidence containing a coarse redundancy that disappears as soon as either of its members is removed.

Definition 15.1 (Channel dependence)**.** Let $\chi : E \to \{+, -\}$ be the coarse evidence polarity. A body of evidence $A \subseteq E$ is *channel-dependent* when

$$\exists x, y \in A \quad x \neq y \quad \text{and} \quad \chi(x) = \chi(y).$$

It is channel-independent when no such pair exists. A *channel circuit* is a channel-dependent set with no proper channel-dependent subset.

The definition is equivalent to the expected closure formulation.

Proposition 15.2 (Closure witness for dependence)**.** *For every $A \subseteq E$,*

$$A \text{ is channel-dependent} \iff \exists x \in A: \; x \in \mathrm{cl}_\chi(A \setminus \{x\}).$$

Thus coarse dependence means that at least one retained atom is already generated by the remainder of the evidence body. STATUS: LEAN-PROVED IN `PEL4/MatroidEvidenceCircuits.lean`.

Theorem 15.3 (Complete circuit classification)**.** *A body $C \subseteq E$ is a channel circuit if and only if there exist distinct $x, y \in E$ such that*

$$\chi(x) = \chi(y) \qquad \text{and} \qquad C = \{x, y\}.$$

Equivalently, the circuits of the coarse 4-PEL partition matroid are exactly the two-element same-polarity pairs. STATUS: LEAN-PROVED IN `PEL4/MatroidEvidenceCircuits.lean`.

Proof sketch. Every distinct same-polarity pair is dependent. Any dependent subset of that pair must contain two distinct elements, so it must contain both members and therefore equals the pair. Conversely, if C is minimally dependent, choose two distinct same-polarity witnesses $x, y \in C$. The pair $\{x, y\}$ is already dependent, hence minimality forces every member of C to lie in that pair. Since both witnesses lie in C, equality follows. ☐

This theorem turns the informal phrase "redundant evidence" into an exact structural claim. At the coarse bilateral projection, a circuit does not mean that one item of evidence is false, weak, or dispensable in every epistemic sense. It means only that two distinct atoms occupy the same one-bit direction seen by the FDE kernel.

Proposition 15.4 (Circuit closure redundancy)**.** *If $x \neq y$ and $\chi(x) = \chi(y)$, then for every $e \in E$,*

$$e \in \mathrm{cl}_\chi(\{x, y\}) \iff e \in \mathrm{cl}_\chi(\{x\}).$$

By symmetry the same holds with $\{y\}$. Hence removing either member of a circuit leaves the coarse generated channel unchanged. STATUS: LEAN-PROVED.

The circuit-elimination behaviour is correspondingly transparent. Suppose $\{x, y\}$ and $\{x, z\}$ are distinct circuits sharing x. Then all three atoms have the same polarity; if $y \neq z$, the pair $\{y, z\}$ is itself a circuit contained in the union after removing the common element. This is the two-channel instance of the standard circuit-elimination pattern [9].

Proposition 15.5 (Two-channel circuit elimination)**.** *Let x, y, z be pairwise distinct with $\chi(x) = \chi(y) = \chi(z)$. Then*

$$\{x, y\}, \{x, z\} \text{ circuits} \implies \{y, z\} \text{ is a circuit}.$$

STATUS: LEAN-PROVED.

15.1 Contraction as accepted background evidence

Deletion and contraction are the canonical minor operations of matroid theory. The current repository interface, however, presents only a closure operator and has no explicit ground-set field. A full minor API would therefore be premature: removing an atom from the carrier cannot yet be represented without changing the ambient type or extending `MatroidClosure`. Gate 3 instead records the part of contraction semantics that is expressible without pretending that this missing structure already exists.

Fix an evidence atom c and treat it as accepted background evidence. Define

$$\mathrm{cl}_\chi^{/c}(A) := \mathrm{cl}_\chi(A \cup \{c\}).$$

This is a contraction-style background closure rather than a packaged matroid minor.

Proposition 15.6 (Parallel evidence becomes loop-like under background)**.** *For any atom e,*

$$\chi(c) = \chi(e) \quad \Longrightarrow \quad e \in \mathrm{cl}_{\chi}^{/c}(\varnothing),$$

whereas

$$\chi(c) \neq \chi(e) \quad \Longrightarrow \quad e \notin \mathrm{cl}_{\chi}^{/c}(\varnothing).$$

Thus accepting one positive atom as background makes every positive parallel atom generated without further evidence, while negative atoms remain a separate information direction; and conversely for negative background evidence. Status: Lean-proved.

This is the strongest epistemic reading justified by the present formalism. In a genuine contraction of a partition matroid, the parallel partners of the contracted element become loops. Here the formal result captures exactly the corresponding closure fact: once one channel representative is fixed in the background, additional same-channel atoms contribute no new coarse rank.

Gate-3 boundary. The classification is exact only after the positive/negative projection. At a finer level, two positive reports may differ in reliability, causal origin, probability mass, inferential route, or independence of source. Such information can split a coarse two-element circuit into an independent fine-grained set. Consequently, "minimal evidential redundancy" is always indexed to a chosen projection.

Next structural question. Gate 4 should promote the lightweight closure presentation to an explicit finite-ground-set matroid layer, or bridge the project to a standard matroid API. Only then can deletion and contraction be stated as actual minors rather than background-closure probes. That gate would allow a precise comparison between source removal, background acceptance, rank loss, loop creation, and preservation of the four-valued state.

16 Genuine minors and residual evidence capacity

Gate 3 could express contraction only as a background-closure probe because the lightweight `MatroidClosure` interface had no explicit ground set. Gate 4 removes that limitation. The formal layer now records a ground predicate together with closure, ground containment, extensivity, monotonicity, idempotence, and exchange, plus a finite list certifying that every ground atom occurs in a finite support. The implementation is deliberately local to the repository and does not add Mathlib as a dependency.

Definition 16.1 (Explicit-ground deletion and contraction)**.** Let M be a ground-set matroid closure on E, with ground set $E(M)$, and let $D, C \subseteq E(M)$. Gate 4 defines

$$E(M \setminus D) = E(M) \setminus D, \qquad \mathrm{cl}_{M \setminus D}(A) = \mathrm{cl}_M(A) \setminus D,$$

and

$$E(M/C) = E(M) \setminus C, \qquad \mathrm{cl}_{M/C}(A) = \mathrm{cl}_M(A \cup C) \setminus C.$$

These are the standard closure formulas for deletion and contraction.

Theorem 16.2 (Minor closure preservation)**.** *Both constructions again satisfy the explicit-ground closure axioms, including Mac Lane–Steinitz exchange. The finite-ground wrappers retain a finite support certificate for the smaller ground set.* Status: Lean-proved in `PEL4/MatroidEvidenceMinors.lean`; both constructors are axiom-free.

The explicit carrier makes the distinction between the two minor operations mathematically literal. Deletion removes atoms from the problem. Contraction first allows the contracted evidence to participate in generation and then removes it from the residual carrier. It therefore represents a quotient by a fixed structural direction, not the same operation as merely forgetting a source.

16.1 Deletion: redundancy versus dispensability

For the finite two-channel 4-PEL matroid, let x and y be distinct same-polarity atoms in the ground set. Since $\{x, y\}$ is a circuit, deleting x leaves y as a representative of the same channel.

Proposition 16.3 (Parallel deletion preserves the coarse state)**.** *If $x \neq y$, $\chi(x) = \chi(y)$, and $y \in E(M)$, then deleting x preserves the presence or absence of both coarse support channels. Consequently, for every FDE value v,*

$$\mathrm{RealizesFDE}(E(M) \setminus \{x\}, v) \iff \mathrm{RealizesFDE}(E(M), v).$$

STATUS: LEAN-PROVED. THE SUPPORT AND PACKAGED FDE EQUIVALENCES USE ONLY THE REPOSITORY'S ALLOWED STANDARD LEAN AXIOMS.

This sharpens the Gate-3 notion of redundancy. Circuit membership implies that a particular atom can be removed without changing the coarse observation only while a parallel representative remains. Redundant relative to closure is therefore not the same as globally dispensable under arbitrary subsequent deletions.

The complementary case is exact as well. Call c a unique channel representative when it is the only ground atom of polarity $\chi(c)$.

Proposition 16.4 (Unique-representative deletion profile)**.** *Deleting a unique channel representative removes exactly its own support channel; every different polarity channel is preserved. In the bilateral four-valued projection, the only possible one-atom information losses are therefore*

$$\mathbf{T} \longrightarrow \mathbf{N}, \qquad \mathbf{F} \longrightarrow \mathbf{N}, \qquad \mathbf{B} \longrightarrow \mathbf{T}, \qquad \mathbf{B} \longrightarrow \mathbf{F},$$

with the direction determined by the deleted polarity and by which opposite channel was present beforehand. STATUS: THE EXACT CHANNEL-PROFILE STATEMENT IS LEAN-PROVED AND AXIOM-FREE; THE DISPLAYED FOUR-VALUE ARROWS ARE ITS IMMEDIATE BILATERAL COROLLARIES.

Deletion thus exposes a useful distinction:

$$\text{coarse redundancy} \neq \text{unconditional removability}.$$

An atom in a circuit is removable relative to another representative, but the last representative of a channel is structurally pivotal for the FDE projection.

16.2 Contraction: quotienting an independent direction

The canonical finite channel matroid has no loops before taking minors. For a single contracted atom c, the general contraction identity gives

$$e \text{ is a loop in } M/c \iff e \in E(M),\ e \neq c, \text{ and } e \in \mathrm{cl}_M(\{c\}).$$

For the two-channel closure this becomes a complete polarity classification.

Theorem 16.5 (Contraction loop profile)**.** *For every remaining atom e,*

$$e \textit{ is a loop in } M/c \iff e \in E(M), \quad e \neq c, \quad \chi(e) = \chi(c).$$

Thus contracting one positive atom turns every remaining positive parallel atom into a loop, while no negative atom becomes a loop; conversely for negative contraction. STATUS: LEAN-PROVED AND AXIOM-FREE IN `PEL4/MatroidEvidenceMinors.lean`.

This motivates a second support notion. Ordinary coarse support asks only whether a polarity-labelled atom remains in the ground set. Define *nonloop support* to require a remaining ground atom of that polarity that is not a loop. Before minor operations these notions coincide because the canonical channel matroid is loopless.

Proposition 16.6 (Residual channel capacity under contraction)**.** *After contracting c:*

(i) the channel $\chi(c)$ has no nonloop support;

(ii) every channel different from $\chi(c)$ has nonloop support after contraction exactly when it had ordinary ground support before contraction.

Hence contraction quotients out exactly one coarse independent direction and leaves the opposite independent direction active when present. Status: Lean-proved and axiom-free in `PEL4/MatroidEvidenceMinorSemantics.lean`.

The distinction between ground support and nonloop support matters. A remaining same-polarity atom may still be physically present in the contracted carrier while contributing zero residual rank because it has become a loop. Consequently the Gate-1 channel-rank reading as "number of present support bits" is exact for the original loopless channel matroid but is not automatically invariant under minors. After contraction, polarity occupancy and residual independent-information capacity must be tracked separately.

Epistemic boundary. Genuine matroid contraction is a structural quotient. It is not, by itself, AGM belief contraction, Bayesian conditioning, public announcement, source acceptance, or a memory-bearing epistemic update. In particular, the contracted atom is absent from the minor ground set. If an epistemic model is meant to remember that atom as accepted background evidence, that memory must be represented by an additional state component rather than silently identified with the matroid minor. Gate 4 therefore replaces the Gate-3 metaphor with a formal separation:

$$\text{source deletion} \neq \text{structural contraction} \neq \text{remembered background acceptance.}$$

Next structural question. The finite-ground minor layer now makes it possible to ask whether the six-cell reliability refinement admits a fine-grained matroid whose deletion and contraction project homomorphically to the two-channel minors. That question would test whether source reliability merely refines parallel classes or destroys the coarse minor correspondence altogether.

17 Classical recovery from evidential regularity

The preceding matroid analysis raises a natural reverse question. Four-valued semantics permits both evidential gaps and evidential gluts; must classical logic therefore be rejected globally, or can a classical fragment be recovered when the underlying evidence structure satisfies stronger regularity conditions? This section formalizes the latter possibility at the coarse two-channel level.

The result is deliberately local and structural. I do not assume bivalence as an extra semantic postulate. Instead, the positive and negative evidence channels are constrained, and the induced four-valued state is then classified. The resulting recovery separates three ideas that are easily conflated: local classical value, recovery of individual classical inference principles, and persistence of classicality under modal or evidential change.

17.1 Completeness and consistency of the channel geometry

Let A be a body of evidence with polarity map $\chi : E \to \{+, -\}$. Write

$$S_+(A) \quad \text{and} \quad S_-(A)$$

for support of the positive and negative channels. Define

$$\begin{aligned}\mathsf{Complete}(A) &:\Longleftrightarrow S_+(A) \vee S_-(A),\\ \mathsf{Consistent}(A) &:\Longleftrightarrow \neg(S_+(A) \wedge S_-(A)),\\ \mathsf{Regular}(A) &:\Longleftrightarrow \mathsf{Complete}(A) \wedge \mathsf{Consistent}(A).\end{aligned}$$

At the value level let $\mathsf{GapFree}(v)$ mean that $v \neq \mathbf{N}$ in the bilateral sense, and $\mathsf{GlutFree}(v)$ that $v \neq \mathbf{B}$ in the bilateral sense. The Lean definitions use the already existing Boolean gap/glut tests rather than introducing a second four-valued algebra.

Theorem 17.1 (Geometry–value recovery)**.** *Suppose A realizes the FDE value v. Then*

$$\begin{aligned}\mathsf{GapFree}(v) &\Longleftrightarrow \mathsf{Complete}(A),\\ \mathsf{GlutFree}(v) &\Longleftrightarrow \mathsf{Consistent}(A),\\ \mathsf{Regular}(A) &\Longleftrightarrow v \in \{\mathbf{T}, \mathbf{F}\}.\end{aligned}$$

Status: Proved; trust inventory in Appendix A.

Thus the classical two-valued slice need not be imposed by deleting $\mathbf{N}$ and $\mathbf{B}$ from the semantics. It is exactly the image of evidence bodies satisfying both coverage and non-conflict at the coarse channel level:

$$\boxed{\mathsf{Complete} + \mathsf{Consistent} \quad \Longleftrightarrow \quad \{\mathbf{T}, \mathbf{F}\}.}$$

This is a recovery theorem inside the existing four-valued semantics, not a claim that all evidence bodies satisfy its hypotheses.

17.2 The recovered propositional slice

The recovered values are closed under the existing FDE propositional operations.

Proposition 17.2 (Classical subalgebra at the value level)**.** *If $v, w \in \{\mathbf{T}, \mathbf{F}\}$, then*

$$\neg v \in \{\mathbf{T}, \mathbf{F}\}, \qquad v \wedge w \in \{\mathbf{T}, \mathbf{F}\}, \qquad v \vee w \in \{\mathbf{T}, \mathbf{F}\}.$$

Status: Proved; trust inventory in Appendix A.

The proposition is modest but important. Recovery is not merely a pointwise classification of two values: the existing four-valued connectives restrict to a closed classical propositional fragment on that slice.

17.3 LEM and EFQ return for different reasons

The two most conspicuous departures from classical reasoning separate cleanly under the bilateral semantics. Let

$$\mathrm{LEM}(v) = v \vee \neg v, \qquad \mathrm{Contr}(v) = v \wedge \neg v.$$

For LP-style designation, a value is tolerated when its positive bit is present.

Proposition 17.3 (Separated recovery conditions)**.** *For every FDE value v:*

(i) $\mathrm{LEM}(v)$ *is LP-designated exactly when v is gap-free;*

(ii) $\mathrm{LEM}(v) = \mathbf{T}$ *exactly when $v \in \{\mathbf{T}, \mathbf{F}\}$;*

(iii) if v is glut-free, then $\mathrm{Contr}(v)$ *LP-entails every conclusion value.*

Status: Proved; trust inventory in Appendix A.

Consequently, excluded middle and explosion do not reappear as an inseparable "classical package." Gap-freedom is the relevant structural boundary for tolerant excluded middle, whereas glut-freedom is the relevant boundary for LP ex falso. Full local classicality requires both. In schematic form,

$$\begin{aligned}\text{no gap} &\implies \text{tolerant LEM},\\ \text{no glut} &\implies \text{LP EFQ},\\ \text{no gap} + \text{no glut} &\iff \mathbf{T}/\mathbf{F}\ \text{slice}.\end{aligned}$$

The LP-explosion statement is vacuous in the precise semantic sense that, under GlutFree, the contradiction premise is not designated. It should therefore not be read as a proof that an inconsistent evidential state has somehow generated arbitrary content.

17.4 Stability is persistence, not local bivalence

The modal-probabilistic refinement already distinguishes local status from stability through an accessibility range. This distinction survives the recovery analysis. Local completeness and consistency are sufficient for a local **T** or **F** value; no modal stability assumption is needed for that classification.

Proposition 17.4 (Stable propagation of recovered classicality)**.** *Let $v : W \to \{\mathbf{N}, \mathbf{T}, \mathbf{F}, \mathbf{B}\}$. If v is stable at w and $v(w)$ is evidentially regular, then every world accessible from w carries a classical value. In fact stability gives the same classical value throughout the accessible range.* Status: Proved; trust inventory in Appendix A.

This motivates a three-step reading of the original hypothesis:

$$\boxed{\text{coverage} \to \text{no gap}, \qquad \text{consistency} \to \text{no glut}, \qquad \text{stability} \to \text{persistence.}}$$

The third condition strengthens classicality dynamically or modally rather than creating local bivalence in the first place.

17.5 Deletion robustness and fragility

Gate 4 makes it possible to ask whether recovered classicality survives structural changes in its evidence base. The answer already distinguishes redundancy from fragility.

Theorem 17.5 (Parallel deletion preserves regularity)**.** *Suppose $x \neq y$ are evidence atoms of the same polarity and y remains in the ground set. Deleting x preserves channel completeness, channel consistency, and therefore channel regularity exactly:*

$$\mathsf{Regular}(E \setminus \{x\}) \iff \mathsf{Regular}(E).$$

Status: Proved; trust inventory in Appendix A.

Thus local classical recovery is robust to deletion of a redundant parallel source. By contrast, the last source carrying a classical channel is structurally critical.

Theorem 17.6 (Unique-source deletion destroys completeness)**.** *Suppose E is channel-regular and c is the unique representative of its polarity. Then*

$$\neg\mathsf{Complete}(E \setminus \{c\}).$$

Moreover, every FDE value realized by the remaining evidence is **N**. Status: Proved; trust inventory in Appendix A.

This theorem gives a precise sense in which a classical state can be genuine but fragile. The state is classical because the current evidence geometry is complete and consistent, yet removal of a structurally indispensable source reopens an evidential gap:

$$\mathbf{T} \to \mathbf{N} \qquad \text{or} \qquad \mathbf{F} \to \mathbf{N}.$$

Classicality is therefore not merely a label on a value; its robustness depends on redundancy in the supporting evidence geometry.

17.6 Independent-capacity recovery and contraction

Gate 4 also showed that ordinary ground occupancy and residual independent-information capacity diverge after contraction. This motivates a stronger recovery predicate. Call a channel *nonloop-supported* when it has a remaining ground atom that is not a loop in the current matroid minor. Define NLComplete, NLConsistent, and NLRegular by replacing ordinary channel support with nonloop support in the preceding definitions.

Theorem 17.7 (Occupancy and capacity coincide before minors). *For the original finite channel matroid,*

$$\mathsf{NLRegular}(M) = \mathsf{Regular}(E).$$

Equivalently, nonloop completeness and consistency coincide separately with ordinary channel completeness and consistency. Status: Proved; trust inventory in Appendix A.

This equivalence holds because the canonical finite channel matroid is loopless before minors. Hence the original $\mathbf{T}/\mathbf{F}$ recovery theorem can be read either as a statement about occupancy or about independent channel capacity.

Contraction breaks that identification.

Theorem 17.8 (Contraction destroys independent classical capacity). *Let E be channel-regular and let $c \in E$. After contracting c, the resulting minor is not nonloop-complete and therefore not nonloop-regular:*

$$\neg\mathsf{NLComplete}(M/c), \qquad \neg\mathsf{NLRegular}(M/c).$$

Status: Proved; trust inventory in Appendix A.

The reason is structural. Contraction makes every remaining atom of c's polarity a loop, while channel regularity already excluded support for the opposite polarity. Thus no independent channel direction remains.

The distinction is observable even when a same-polarity atom survives in the ground set.

Proposition 17.9 (Occupancy without capacity). *Suppose $c \neq y$ are ground atoms of the same polarity. After contracting c, that polarity is still ordinarily supported by y, but it has no nonloop support.* Status: Proved; trust inventory in Appendix A.

Therefore a coarse support reading can continue to display a channel that has become structurally informationally inert. Gate 5 consequently distinguishes

$$\boxed{\text{classical-valued occupancy} \quad \neq \quad \text{classicality sustained by residual independent capacity.}}$$

This is the first formal refinement of "informational stability" in the recovery program: classical value recovery is local, whereas structural classicality must also track what independent information survives admissible transformations.

17.7 Scope and next boundary

The current Gate 5 results establish value-level, modal-persistence, deletion, and independent-capacity recovery layers. They do *not* yet prove completeness of a classical proof calculus, classicality of every modal formula, or preservation under every evidence update. In particular, the present LP ex-falso theorem is a value-level semantic recovery under glut-freedom; its premise is undesignated there, and no stronger proof-theoretic conclusion is inferred.

The reliability refinement raises the next boundary. A coarse state can be regular while distinct fine reliability profiles project to it. A later gate should ask whether classical recovery commutes with forgetting reliability and whether a fine-grained notion of independent-capacity regularity survives that projection. Likewise, a formula-level recovery theorem should quantify regularity over the atomic valuation and prove closure inductively through the language, treating modal operators separately from the already verified propositional $\mathbf{T}/\mathbf{F}$ subalgebra.

18 Formula-level classical recovery and operator boundaries

Gate 5 isolated the classical value sector by structural conditions on coarse evidence. A separate question is whether that sector is compositional at the object-language level. This section answers the propositional part positively, identifies a sharp boundary at Lockean belief, and proves preservation for the primitive knowledge and possibility transports on classical accessible profiles.

The distinction is important. From

$$v \in \{\mathbf{T}, \mathbf{F}\}$$

it does not follow merely by terminology that every compound expression evaluated from v is classical. The preservation property must be checked constructor by constructor.

18.1 Structural induction on the propositional fragment

Let `Formula.IsPropositional` denote the fragment generated by atoms, negation, and conjunction, excluding the Lockean belief constructor. The existing OR and material-implication forms are syntactic macros built from those constructors and therefore remain inside the fragment.

At a world w, write $\mathsf{AtomicClassicalAt}(m, w)$ when every atomic valuation belongs to $\{\mathbf{T}, \mathbf{F}\}$.

Theorem 18.1 (Propositional formula recovery)**.** *If every atom is classical at w and φ is propositional, then*

$$[\![\varphi]\!]_{m,w} \in \{\mathbf{T}, \mathbf{F}\}.$$

Status: Proved; trust inventory in Appendix A.

The proof is structural induction. The atomic case is the hypothesis; the negation and conjunction steps use the Gate 5 closure theorems for the recovered classical slice. No second truth-value algebra is introduced.

Two familiar classical laws therefore lift from values to formulas on this restricted sector.

Corollary 18.2 (Formula-level excluded middle)**.** *Under the hypotheses of Theorem 18.1,*

$$[\![\varphi \vee \neg\varphi]\!]_{m,w} = \mathbf{T}.$$

Status: Proved; trust inventory in Appendix A.

Corollary 18.3 (Restricted formula-level explosion)**.** *Under the same hypotheses,*

$$[\![\varphi \wedge \neg\varphi]\!]_{m,w} = \mathbf{F},$$

so the contradiction value LP-validly implies any conclusion value at that model/world. This is a recovery theorem relative to regular propositional valuations, not unrestricted LP entailment in arbitrary 4-PEL models. Status: Proved; trust inventory in Appendix A.

18.2 Lockean belief can reopen a gap

Atomic bivalence is insufficient once the probabilistic belief constructor is admitted. Gate 6 contains a two-world witness whose only atomic values are **T** and **F**. The accessible profile splits the positive and negative extensions evenly, with mass 1/2 on each side and threshold 3/5. Therefore neither threshold is reached:

$$\mathsf{B}p = \mathbf{N}.$$

Proposition 18.4 (Classical atoms do not force classical belief)**.** *There exists a finite 4-PEL model whose atomic valuation is classical at every world but in which a belief formula evaluates to* **N**. Status: Finite-witness; trust inventory in Appendix A.

The witness separates two kinds of nonclassicality. No contradictory support is needed. The belief gap is generated by probabilistic indecision:

$$P^{+}(p) < c \qquad \text{and} \qquad P^{-}(p) < c.$$

In the concrete witness both inequalities are $1/2 < 3/5$.

18.3 The exact local belief-recovery condition

To expose this boundary, Gate 6 names the positive and negative threshold decisions made by the existing belief implementation. Write b_+ and b_- for these Boolean decisions. Define

$$\begin{aligned} \mathsf{BelComplete} &:\Longleftrightarrow b_+ \vee b_-, \\ \mathsf{BelConsistent} &:\Longleftrightarrow \neg(b_+ \wedge b_-), \\ \mathsf{BelRegular} &:\Longleftrightarrow \mathsf{BelComplete} \wedge \mathsf{BelConsistent}. \end{aligned}$$

Theorem 18.5 (Exact Lockean recovery boundary)**.** *For every model, agent, world, and FDE-valued input profile,*

$$\mathsf{B}(v) \in \{\mathbf{T}, \mathbf{F}\} \quad \Longleftrightarrow \quad \mathsf{BelRegular}(v).$$

Equivalently, exactly one of the two threshold decisions must fire. Status: Proved; trust inventory in Appendix A.

The balanced Gate 6 witness is formally classified as threshold-incomplete but threshold-consistent. Thus its failure of classicality is specifically a failure of coverage at the belief threshold, not a glut.

Theorem 18.5 is exact but deliberately local. It does not yet derive threshold consistency from finite additivity plus $c > 1/2$. Such a derivation requires the stronger `FiniteProbability Integrity` layer together with a proof that the positive and negative events form the relevant complementary partition. The lightweight legacy `Model` interface is too weak for that step, and no stronger probabilistic conclusion is inferred from it here.

18.4 Knowledge and possibility preserve classical profiles

The conservative modal language contains primitive evidence-stable knowledge and raw accessibility possibility. Unlike Lockean belief, both preserve the recovered classical sector when their accessible semantic input profile is already classical.

Theorem 18.6 (Modal transport preservation)**.** *Let $v : W \to \{\mathbf{N}, \mathbf{T}, \mathbf{F}, \mathbf{B}\}$. If every world accessible from w has $v(u) \in \{\mathbf{T}, \mathbf{F}\}$, then both*

$$\mathsf{Poss}_i(v)(w) \quad \textit{and} \quad \mathsf{K}_i(v)(w)$$

belong to $\{\mathbf{T}, \mathbf{F}\}$. Status: Proved; trust inventory in Appendix A.

For raw possibility, classicality turns the negative bit into the Boolean complement of the positive bit across the finite accessibility list. For evidence-stable knowledge, an unstable profile collapses to $\mathbf{F}$, while a stable classical profile retains a classical result. The theorem also covers the repository's empty accessibility conventions.

This yields a stronger compositional statement for the modal language.

Theorem 18.7 (Belief-free modal recovery)**.** *Suppose every atomic value at every world is classical. Then every modal formula built from atoms, negation, conjunction, primitive knowledge, and primitive raw possibility, but containing no Lockean belief constructor, evaluates in* $\{\mathbf{T}, \mathbf{F}\}$ *at every world.* Status: Proved; trust inventory in Appendix A.

Hence the first operator boundary of formula-level classical recovery is not "modality" in general. It is specifically the thresholding operation used by Lockean belief.

18.5 A recursive recovery contract for the full legacy syntax

Gate 6 packages the preceding result into a semantic admissibility predicate `Formula.ClassicalRecoveryAdmissibleAt`. Atoms must be classical; negation and conjunction recurse without new obligations. A belief node requires two things: its subformula must remain recovery-admissible throughout the accessible range, and the local belief threshold pair must be regular.

Theorem 18.8 (Full legacy-formula recovery under admissibility)**.** *For every legacy formula* φ*,*

$$\mathsf{ClassicalRecoveryAdmissibleAt}(m, w, \varphi) \quad \Longrightarrow \quad [\![\varphi]\!]_{m,w} \in \{\mathbf{T}, \mathbf{F}\}.$$

Status: Proved; trust inventory in Appendix A.

This is the first formula-level version of the classical-recovery hypothesis. It should not be read as an unrestricted theorem that all 4-PEL formulas are classical. Instead it identifies the constructor-local obligations required for classicality to survive composition.

18.6 Relation to the existing CPEL translation

The repository already defines positive and negative translations `tr_pos` and `tr_neg` into a classical probabilistic epistemic syntax. The accompanying source comment sketches a completeness reduction, but no Lean theorem presently proves that reduction. Gate 6 therefore does not use that comment as evidence for completeness.

A subsequent step can compare the direct recovery theorem above with the split translation and ask whether regular valuations force a collapse or correspondence between the positive and negative classical translations. Until such a theorem is proved, the mechanized result is the direct semantic recovery developed here.

Gate boundary. Gate 6 establishes compositional recovery and isolates the Lockean threshold as a special source of nonclassicality. The probability-integrity derivation deliberately left open here is discharged in Section 19; the proof-system or CPEL-translation completeness question remains open.

19 Lockean threshold phase transition under probability integrity

Gate 6 isolated an exact local condition for classical Lockean belief, but its consistency component was still imposed semantically. This section derives that component from the stronger finite-probability layer and then separates the role of the threshold itself from the legacy model contract.

The result is a sharper boundary. In an integrity-certified model with a classical accessible profile, positive and negative support are not merely two unrelated event lists. They form a complementary partition of the accessibility support. The legacy supermajority threshold $c > 1/2$ therefore excludes belief gluts structurally. The only remaining failure mode for classical belief is indecision.

19.1 Classical profiles induce complementary support masses

Fix an agent i, a world w, and an FDE-valued profile v on the worlds accessible from w. Let

$$E^+ = \{u \in R_i(w) : v(u).\mathsf{pos} = \mathsf{true}\}, \qquad E^- = \{u \in R_i(w) : v(u).\mathsf{neg} = \mathsf{true}\}.$$

If every accessible value is classical, then each accessible world lies in exactly one of these two events. Hence they are disjoint and cover the accessibility support.

Theorem 19.1 (Complementary classical support masses)**.** *Let m satisfy `ModelProbabilityIntegrity`. If every value in the accessible profile is in $\{\mathbf{T}, \mathbf{F}\}$, then*

$$\mu_{i,w}(E^+) + \mu_{i,w}(E^-) = 1.$$

Status: Proved; trust inventory in Appendix A.

The proof uses the finite-probability integrity contract rather than the lightweight legacy probability interface. Duplicate-freedom of the accessibility support is transported through the two filters, the events are shown disjoint, finite additivity combines their masses, and extensionality identifies their union with the whole support.

This theorem closes the precise gap left open at the end of Gate 6: once the input profile is classical, the two threshold masses are genuine complements.

19.2 Supermajority eliminates the glut branch

Write

$$p = P^+(v), \qquad n = P^-(v).$$

For a classical accessible profile under probability integrity, Theorem 19.1 gives $p + n = 1$. The existing 4-PEL model contract also requires

$$c > \frac{1}{2}.$$

These two facts are enough to rule out simultaneous threshold acceptance.

Theorem 19.2 (Supermajority threshold consistency)**.** *If $p + n = 1$ and $c > 1/2$, then*

$$\neg(c \le p \wedge c \le n).$$

Consequently, every integrity-certified classical accessible profile satisfies `BeliefThresholdConsistent`. Status: Proved; trust inventory in Appendix A.

Thus the exact Gate 6 recovery boundary simplifies in the stronger model class.

Corollary 19.3 (Classical belief iff decisive belief)**.** *Under `ModelProbabilityIntegrity` and a classical accessible profile,*

$$\mathsf{B}(v) \in \{\mathbf{T}, \mathbf{F}\} \iff \mathsf{BelComplete}(v).$$

Equivalently,

$$\mathsf{B}(v) \notin \{\mathbf{T}, \mathbf{F}\} \iff \neg\mathsf{BelComplete}(v).$$

Status: Proved; trust inventory in Appendix A.

The second form is conceptually useful. Once the profile is classical and the probability laws are certified, nonclassical Lockean belief is exactly a gap caused by indecision. A **B**-valued belief is no longer a live possibility in this sector.

19.3 The model-independent threshold boundary

The preceding theorem uses the existing model's built-in supermajority assumption. To see what role $1/2$ plays independently of that design choice, Gate 7 also factors the threshold arithmetic into a model-free layer. For rational masses p, n with

$$p + n = 1,$$

define threshold completeness by $c \le p \vee c \le n$, and threshold consistency by $\neg(c \le p \wedge c \le n)$.

Two complementary one-sided results follow.

Theorem 19.4 (Threshold phase boundary)**.** *For complementary masses $p + n = 1$:*

(i) if $c \le 1/2$, threshold completeness is guaranteed, so a gap is impossible;

(ii) if $c > 1/2$, threshold consistency is guaranteed, so a glut is impossible.

Status: Proved; trust inventory in Appendix A.

The two implications make the remaining obstruction regime-dependent rather than merely one-sided.

Corollary 19.5 (Exact regime reduction)**.** *For $p + n = 1$,*

$$c \le \frac{1}{2} \quad \Longrightarrow \quad \mathsf{ThresholdRegular} \Longleftrightarrow \mathsf{ThresholdConsistent},$$

whereas

$$c > \frac{1}{2} \quad \Longrightarrow \quad \mathsf{ThresholdRegular} \Longleftrightarrow \mathsf{ThresholdComplete}.$$

Status: Proved; trust inventory in Appendix A.

Thus the critical value separates which half of regularity still needs to be checked. At or below one half, underdecision is ruled out and only overacceptance can spoil regularity. Above one half, overacceptance is ruled out and only underdecision can spoil regularity.

The critical value is therefore not merely a conventional midpoint. It separates two qualitatively different failure guarantees. Lower thresholds protect against indecision but permit overacceptance; strict supermajority thresholds protect against overacceptance but permit indecision.

The inclusivity of the existing comparison matters at the boundary. Because belief uses $\ge c$, an exact tie behaves as follows.

Theorem 19.6 (Exact tie dichotomy)**.** *For $p = n = 1/2$:*

$$c \le \frac{1}{2} \quad \Longrightarrow \quad c \le p \wedge c \le n,$$

whereas

$$c > \frac{1}{2} \quad \Longrightarrow \quad \neg(c \le p) \wedge \neg(c \le n).$$

In particular, at $c = 1/2$ the tie is glut-like rather than gap-like. Status: Proved; trust inventory in Appendix A.

In four-valued language, the threshold image of the tie jumps across the critical value:

$$\boxed{c \le \frac{1}{2} : \mathbf{B} \qquad | \qquad c > \frac{1}{2} : \mathbf{N}}$$

where the labels describe the two threshold bits, not the underlying world-level truth value.

19.4 No universal inclusive Lockean cutoff

The tie immediately yields a sharp impossibility statement.

Theorem 19.7 (No universal inclusive threshold). *For every rational threshold c, the complementary profile*

$$p = n = \frac{1}{2}$$

is not threshold-regular. Hence no single inclusive Lockean cutoff makes every complementary probability profile classically two-valued. STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

This result should not be read as a defect of four-valued semantics. It is a property of inclusive thresholding on a complementary probability line. A single cutoff cannot simultaneously guarantee both completeness and consistency at the exact midpoint. The legacy 4-PEL choice $c > 1/2$ resolves the ambiguity in favor of consistency, sending the balanced case to the gap side rather than the glut side.

19.5 Formula recovery with one fewer obligation

Gate 6 defined `Formula.ClassicalRecoveryAdmissibleAt`; every belief node required both threshold completeness and threshold consistency. Gate 7 can now weaken this recursive contract for integrity-certified models.

Define `Formula.ProbabilityRecoveryAdmissibleAt` exactly as before for atoms, negation, and conjunction. At a belief node, require only:

(i) recursive admissibility of the subformula throughout the accessible range;

(ii) local threshold completeness.

No independent threshold-consistency premise is included.

Theorem 19.8 (Formula recovery under probability integrity). *Let m satisfy `ModelProbabilityIntegrity`. For every legacy formula φ,*

$$\mathsf{ProbabilityRecoveryAdmissibleAt}(m, w, \varphi) \quad \Longrightarrow \quad [\![\varphi]\!]_{m,w} \in \{\mathbf{T}, \mathbf{F}\}.$$

The corresponding global admissibility predicate yields classical evaluation at every world. STATUS: PROVED; TRUST INVENTORY IN APPENDIX A.

At belief nodes the induction first obtains a classical accessible subformula profile. Probability integrity then supplies complementary masses, the built-in supermajority threshold supplies consistency, and the sole explicit belief premise, completeness, finishes the classical-recovery step.

This gives the Gate 7 hierarchy

$$\text{classical accessible profile} + \text{probability integrity} + c > \tfrac{1}{2} \Longrightarrow \text{threshold consistency},$$

followed by

$$\text{threshold completeness} \Longleftrightarrow \text{classical Lockean belief}.$$

Gate boundary. Gate 7 derives the consistency half of Lockean classical recovery from the stronger finite-probability semantics, formalizes the midpoint phase transition independently of the model, proves the exact tie obstruction for every inclusive threshold, and lifts the reduced recovery contract through the full legacy formula syntax. It does not prove that arbitrary formulas are threshold-complete, nor does it prove the commented CPEL proof-system or translation completeness claim. Decisiveness remains a genuine semantic obligation.

20 Split CPEL representation and the classical anti-diagonal

The preceding recovery gates characterize when a four-valued evaluation returns to $\{\mathbf{T}, \mathbf{F}\}$. Gate 8 asks a different question: what does that recovery look like through the repository's existing split translation into the classical probabilistic epistemic syntax `CPELFormula`?

The translation was already present syntactically. Positive support and negative support are sent to separate Boolean-tagged atomic channels,

$$p^+ \mapsto (p, \mathsf{true}), \qquad p^- \mapsto (p, \mathsf{false}),$$

with negation exchanging the two translations, conjunction preserving positive support conjunctively, and negative support translated through derived classical disjunction. Until this gate, however, the target syntax had no explicit evaluator connected back to the finite 4-PEL model semantics.

20.1 An explicit Boolean target semantics

Gate 8 defines `evalCPEL` on `CPELFormula(Atom×Bool)Ag`. At an atom, the Boolean tag selects either the positive or negative FDE support bit. Classical negation and conjunction use Boolean negation and conjunction. Belief uses the same accessibility list, finite event-mass function, and threshold as the source model, but thresholds the Boolean truth set of the translated subformula.

The derived CPEL disjunction then evaluates as ordinary Boolean disjunction. This small bridge is enough to state the first main result without any recovery assumption.

Theorem 20.1 (Exact split-translation semantics)**.** *For every legacy formula φ, finite 4-PEL model m, and world w,*

$$[\![\mathrm{tr}_+(\varphi)]\!]^C_{m,w} = \left([\![\varphi]\!]_{m,w}\right)^+,$$

and

$$[\![\mathrm{tr}_-(\varphi)]\!]^C_{m,w} = \left([\![\varphi]\!]_{m,w}\right)^-.$$

Status: Proved; trust inventory in Appendix A.

The proof is by structural induction through atoms, negation, conjunction, and Lockean belief. The belief case is important: the induction hypotheses identify the translated Boolean event lists pointwise with the source positive and negative support-event lists, so the same threshold comparison is recovered on both sides. No classicality, probability-integrity, or recovery premise is required.

Package the two target truth values into one FDE-shaped pair,

$$\mathrm{Pair}_C(m, w, \varphi) = \left([\![\mathrm{tr}_+(\varphi)]\!]^C_{m,w}, [\![\mathrm{tr}_-(\varphi)]\!]^C_{m,w}\right).$$

The bitwise theorem immediately yields a pointwise representation theorem.

Theorem 20.2 (Two-Boolean representation)**.** *For every m, w, φ,*

$$\boxed{[\![\varphi]\!]_{m,w} = \mathrm{Pair}_C(m, w, \varphi).}$$

Thus the split target semantics represents all four FDE values, not only the classically recovered ones. Status: Proved; trust inventory in Appendix A.

This theorem should be read semantically and pointwise. It does not identify the source and target proof systems, and it is not a strong-completeness theorem.

20.2 The classical anti-diagonal

Because an FDE value is literally a pair of Boolean support coordinates, its four vertices are

$$\mathbf{N} = (0,0), \qquad \mathbf{T} = (1,0), \qquad \mathbf{F} = (0,1), \qquad \mathbf{B} = (1,1).$$

The classically recovered values are exactly the two vertices on which one coordinate is the Boolean complement of the other.

Theorem 20.3 (Classical anti-diagonal)**.** *For every FDE value v,*

$$v \in \{\mathbf{T}, \mathbf{F}\} \quad \Longleftrightarrow \quad v^- = \neg v^+.$$

Consequently, for every formula,

$$\boxed{[\![\varphi]\!]_{m,w} \in \{\mathbf{T}, \mathbf{F}\} \quad \Longleftrightarrow \quad [\![\mathrm{tr}_-(\varphi)]\!]^C_{m,w} = \neg[\![\mathrm{tr}_+(\varphi)]\!]^C_{m,w}.}$$

Status: Proved; trust inventory in Appendix A.

The first equivalence is independent of formula syntax. The second combines it with the exact split-translation theorem. In the Boolean evidence square, the classical sector is therefore the discrete anti-diagonal

$$\{(1,0),(0,1)\} = \{(p,n) \in \{0,1\}^2 : n = \neg p\}.$$

The two nonclassical vertices fail the same equation in opposite ways: **N** lacks both channels, while **B** contains both.

This gives a precise sense in which classical recovery removes one independent Boolean degree of freedom. Outside the recovered slice, positive and negative support are genuinely separate coordinates. On the anti-diagonal, choosing the positive coordinate fixes the negative coordinate. The word "degree" here refers to discrete Boolean freedom, not vector-space dimension.

20.3 Gate 7 recovery collapses the split representation

Gate 7 supplies a sufficient recursive condition for classicality under `ModelProbabilityIntegrity`. Combining that theorem with the anti-diagonal characterization turns semantic recovery into an explicit collapse result.

Corollary 20.4 (Probability-recovery split collapse)**.** *Let m satisfy `ModelProbabilityIntegrity`, and let φ satisfy `Formula.ProbabilityRecoveryAdmissible`. Then at every world w,*

$$[\![\mathrm{tr}_-(\varphi)]\!]^C_{m,w} = \neg[\![\mathrm{tr}_+(\varphi)]\!]^C_{m,w}.$$

Moreover, the full recovered FDE value can be reconstructed from the positive target bit alone:

$$[\![\varphi]\!]_{m,w} = \mathrm{ofClassicalBool}\left([\![\mathrm{tr}_+(\varphi)]\!]^C_{m,w}\right),$$

where $\mathrm{ofClassicalBool}(1) = \mathbf{T}$ *and* $\mathrm{ofClassicalBool}(0) = \mathbf{F}$. Status: Proved; trust inventory in Appendix A.

The logical shape of Gates 5–8 can now be displayed as a compression chain:

$$\begin{gathered}
\text{regular evidence / recursive recovery} \\
\Downarrow \\
[\![\varphi]\!] \in \{\mathbf{T}, \mathbf{F}\} \\
\Downarrow \\
[\![\mathrm{tr}_-(\varphi)]\!]^C = \neg[\![\mathrm{tr}_+(\varphi)]\!]^C \\
\Downarrow \\
\text{one Boolean coordinate reconstructs the recovered value.}
\end{gathered}$$

20.4 Induced consequence, without a target completeness claim

The representation also transports the two source consequence relations already formalized in the artifact. The target notions used here quantify over *CPEL evaluations induced by the same 4-PEL models*; they are not validity relations over an independently axiomatized class of arbitrary CPEL models.

For LP consequence, designatedness is exactly truth of the positive translation. Thus, for all formulas φ, ψ,

$$\boxed{\varphi \models_{\mathrm{LP}} \psi \quad \Longleftrightarrow \quad \mathrm{tr}_+(\varphi) \models^{\mathrm{ind}}_{C,+} \mathrm{tr}_+(\psi).}$$

For strict-to-tolerant consequence, strict truth of the antecedent is represented by the split profile $(\mathsf{true}, \mathsf{false})$, while the consequent remains positively designated. Hence

$$\boxed{\varphi \models_{\mathrm{ST}} \psi \quad \Longleftrightarrow \quad (\mathrm{tr}_+(\varphi) = \mathsf{true} \,\&\, \mathrm{tr}_-(\varphi) = \mathsf{false}) \models^{\mathrm{ind}}_{C} \mathrm{tr}_+(\psi) = \mathsf{true}.}$$

The exact strict-profile lemma is $[\![\varphi]\!] = \mathbf{T}$ iff its two translated bits are $(\mathsf{true}, \mathsf{false})$. Combining the ST equivalence with the existing source soundness theorem transports every ST derivation soundly into the induced split target semantics.

These are exact semantic bridges for the induced evaluator. They neither define a CPEL proof calculus nor establish soundness or completeness for arbitrary target models. In particular, source derivability is transported only in the sound direction already licensed by the source soundness theorem.

Gate boundary. Gate 8 proves an explicit Boolean semantics for the existing split CPEL target, an unconditional pointwise representation of both FDE support bits, an exact characterization of the classical sector as the complement locus, the collapse to one Boolean coordinate under Gate 7 probability recovery, and exact LP/ST consequence bridges for the induced evaluator. It does *not* formalize a CPEL Hilbert system, arbitrary target-model validity, a canonical model, strong completeness, or decidability. The former source comment suggesting that the split translation by itself guaranteed those results has therefore been narrowed to an intended reduction strategy rather than promoted to a theorem.

21 Conclusion

The case study establishes a sequence of bounded results. First, the stipulated evidence-stable knowledge operator factors through accessible-profile homogeneity; probability-only conditionalization realizes all sixteen $\mathsf{K}(\mathsf{B}p)$ endpoint values across a six-world witness family. This is not a claim that one fixed model realizes every update, nor an empirical theory of knowledge.

Second, signed thresholding reduces quantitative support to two Boolean decisions. On a chosen rational affine segment, straddling gives unique wall-hit parameters and their order determines a sequential midpoint phase. An actual discrete update does not supply this interpolated history. At simultaneous hits the inclusive threshold yields $\mathbf{B}$, which may occur only at the boundary parameter.

Separately, normalized rational world weights construct probability-integrity-certified model paths with affine fixed-event masses. Connecting these paths to the supports of arbitrary probability-sensitive formulas, or generating them by admissible updates, remains unproved. The weakest model interface is not silently promoted to a finite probability space.

Third, six-cell reliability tagging preserves coarse support and convex mixing but loses recoverable reliability under projection. Fine current-world stability holds exactly when coarse stability and local fibre rigidity both hold. The S5-frame witnesses show that frame geometry alone cannot restore discarded information. The tags are supplied semantic data, not a derivation of reliability or a six-valued logical calculus.

Fourth, the bilateral evidence layer supports a structural recovery program. Positive and negative evidence channels induce the coarse two-class matroid, whose circuits and finite-ground minors distinguish occupancy from residual independent capacity. Channel completeness and consistency isolate exactly the classical $\{\mathbf{T}, \mathbf{F}\}$ slice. Formula-level recovery then identifies Lockean belief as the relevant obstruction, while finite probability integrity removes the glut branch above the one-half threshold. The exact 50/50 profile remains a universal obstruction to inclusive threshold regularity.

Finally, the existing split CPEL translation now has an explicit semantic bridge. Every four-valued formula evaluation is represented pointwise by the two Boolean values of its positive and negative translations. Classical recovery is exactly the complement equation between those coordinates, selecting the anti-diagonal $\{(1,0),(0,1)\}$ from the four-valued square. Under probability-integrity recovery, the negative coordinate is determined by the positive one and one Boolean target value reconstructs the complete recovered FDE value. LP and strict-to-tolerant consequence are represented exactly in the evaluator induced by 4-PEL models, with source derivability transported in the sound direction. These are representation, collapse, and induced-consequence theorems, not a completeness reduction for an independently axiomatized target proof system.

The contribution is a reproducible mechanized analysis of preservation, dependence, recovery, and information loss. The declaration-level audits, explicit finite witnesses, and proof sketches identify what is established and under which contract. Before journal submission, the most valuable next steps are a strong certificate for the reachability family, a formula-level model-path bridge, a separately formalized CPEL validity/completeness layer if that reduction is to be pursued, and a theorem-by-theorem comparison with the probability and evidence-logics literature. No claim of completeness, topological generality, or priority is inferred from a successful build.

A Formal Correspondence

This appendix maps the principal mathematical claims in the manuscript to the Lean modules and theorem names that support them. The list is selective rather than exhaustive. All names below are in namespace `PEL4`; the audit files enumerate exact fully qualified targets. A theorem's hypotheses remain part of its claim.

Paper claim	Lean theorem / definition	Module
Four FDE values as bilateral Boolean pairs	`FDEValue`, `FDEValue.T/F/B/N`	`PEL4/FDE.lean`
Signed threshold belief	`belief`	`PEL4/Belief.lean`
Primitive modal evaluator with $\mathsf{B}, \mathsf{K}, \Diamond$	`ModalFormula`, `evalModal`	`PEL4/ModalLanguage.lean`
Knowledge equals stability-filtered belief	`modal_knowledge_as_stability_filtered_belief`	`PEL4/ModalKnowledgeBeliefFactorization.lean`
Safe probability-only update	`ConditionalizationAdmissible`, `conditionalize`	`PEL4/Dynamics.lean`
Probability-free invariance	`evalModal_conditionalize_of_probabilityFree`	`PEL4/ModalDynamicsStability.lean`

Four dynamic stability phases	`modal_knowledge_phase_stable_stable`, `modal_knowledge_phase_stable_unstable`, `modal_knowledge_phase_unstable_stable`, `modal_knowledge_phase_unstable_unstable`	`PEL4/ModalDynamicsPhaseClassification.lean`
Both stability-change directions are realizable	`conditionalization_realizes_both_stability_phase_directions`	`PEL4/ModalDynamicsPhaseClassification.lean`
All sixteen $\mathsf{K}(\mathsf{B}p)$ endpoint transitions	`conditionalization_realizes_every_knowledge_value_transition`	`PEL4/ModalDynamicsReachability.lean`
Creation of a knowledge glut	`conditionalization_can_create_knowledge_glut`	`PEL4/ModalDynamicsReachability.lean`
Glut-to-gap and gap-to-truth transitions	`conditionalization_can_turn_knowledge_glut_into_gap`, `conditionalization_can_turn_knowledge_gap_into_truth`	`PEL4/ModalDynamicsReachability.lean`
Exact belief robustness boundary	`modal_belief_value_eq_iff_threshold_bits_eq`	`PEL4/ModalDynamicsRobustness.lean`
Compositional robust formulas are invariant	`evalModal_conditionalize_of_robust`	`PEL4/ModalDynamicsRobustness.lean`
Robustness strictly extends the probability-free fragment	`diagonal_reachability_belief_is_robust`, `diagonal_reachability_belief_not_probabilityFree`	`PEL4/ModalDynamicsRobustness.lean`
Threshold wall count	`thresholdWallCount`	`PEL4/ModalDynamicsGeometry.lean`
Zero wall iff equal FDE state	`thresholdWallCount_eq_zero_iff`	`PEL4/ModalDynamicsGeometry.lean`
Two walls iff both coordinates flip	`thresholdWallCount_eq_two_iff`	`PEL4/ModalDynamicsGeometry.lean`
Wall flip iff endpoint straddling	`threshold_decision_ne_iff_straddles`	`PEL4/ModalDynamicsThresholdCrossing.lean`
Two walls iff both supports straddle	`conditionalization_belief_two_walls_iff_both_supports_straddle`	`PEL4/ModalDynamicsThresholdCrossing.lean`
Affine path	`affineRatPath`	`PEL4/ModalDynamicsAffineCrossing.lean`
Straddling gives unit-interval threshold hit	`thresholdStraddles_has_affine_unit_crossing`	`PEL4/ModalDynamicsAffineCrossing.lean`
Every belief change has an affine wall hit	`conditionalization_belief_change_has_affine_threshold_crossing`	`PEL4/ModalDynamicsAffineCrossing.lean`
Two-wall transition has two affine hits	`conditionalization_belief_two_walls_has_two_affine_crossings`	`PEL4/ModalDynamicsAffineCrossing.lean`
Unique affine threshold time	`affineRatPath_threshold_hit_unique`	`PEL4/ModalDynamicsCrossingOrder.lean`

Crossing-order trichotomy	`affineCrossingPair_order_trichotomy`	`PEL4/ModalDynamicsCrossingOrder.lean`
Simultaneous iff common (c, c) hit	`affineCrossingPair_simultaneous_iff_common_hit`	`PEL4/ModalDynamicsCrossingOrder.lean`
Conditionalized two-wall order classification	`conditionalization_belief_two_walls_crossing_order`	`PEL4/ModalDynamicsCrossingOrder.lean`
Midpoint lies between sequential crossings	`affineCrossingMidpoint_between_positive_first`, `affineCrossingMidpoint_between_negative_first`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Source side before and target side after crossing	`affine_threshold_side_before_crossing_eq_source`, `affine_threshold_side_after_crossing_eq_target`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Positive-first midpoint state	`affineCrossingPair_positive_first_midpoint_state`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Negative-first midpoint state	`affineCrossingPair_negative_first_midpoint_state`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Intermediate is adjacent to both diagonal endpoints	`positiveFirstIntermediate_adjacent_to_diagonal_endpoints`, `negativeFirstIntermediate_adjacent_to_diagonal_endpoints`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Concrete eight-case diagonal table	`affine_diagonal_intermediate_vertex_table`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Full simultaneous/sequential intermediate-phase theorem	`conditionalization_belief_two_walls_intermediate_phase`	`PEL4/ModalDynamicsIntermediatePhase.lean`
Strong finite-probability contract	`FiniteProbabilityIntegrity`, `StrongProbabilityModel`	`PEL4/FiniteProbabilityIntegrity.lean`
Weight-generated measure satisfies integrity	`weightGeneratedMeasure_integrity`	`PEL4/WeightGeneratedProbability.lean`
Convex interpolation preserves distributions	`convexWeightDistribution`	`PEL4/ConvexProbabilitySimplex.lean`
Every fixed event mass is affine	`weightedEventMass_convex`	`PEL4/ConvexProbabilitySimplex.lean`
Complete strong rational model path	`convexStrongModelAt`	`PEL4/ConvexModelPath.lean`
Worlds, accessibility, valuation, and threshold fixed	`convexStrongModelAt_worlds`, `convexStrongModelAt_accessibility`, `convexStrongModelAt_valuation`, `convexStrongModelAt_threshold`	`PEL4/ConvexModelPath.lean`
Fixed-event mass affine along full model path	`convexStrongModelAt_eventMass`	`PEL4/ConvexModelPath.lean`
Nonempty accessibility and fixed-input K invariance	`PaperReview.accessibility_nonempty`, `PaperReview.knowledge_ignores_probability_for_fixed_values`	`PEL4/PaperReviewBoundaries.lean`

Inclusive common hit is B; simultaneous T/F witness	PaperReview.threshold_intersection_is_B, PaperReview.simultaneous_T_F_has_B_hit	PEL4/PaperReviewBoundaries.lean
Six-cell section and nonrecoverable reliability	ModalProbability.SixCellProbability.coarse_untagged, ModalProbability.SixCellProbability.no_reliability_reconstruction	PEL4/ModalProbability/Refinement.lean
Affine coarse projection	ModalProbability.coarse_mix	PEL4/ModalProbability/Refinement.lean
Local fine/coarse stability boundary	ModalProbability.stable_iff_coarse_and_fibre_rigid	PEL4/ModalProbability/StatusTransport.lean
S5-frame reliability and changing glut-mass witnesses	ModalProbability.probability_profile_projection_hides_instability, ModalProbability.stable_B_does_not_mean_constant_mass	PEL4/ModalProbability/StatusTransport.lean
Coarse two-channel matroid closure	channelMatroid, samePolarity_iff_mem_singletonClosure	PEL4/MatroidEvidence.lean
Topological exchange exactly at singleton symmetry	topologicalClosureExchange_iff_r0Like	PEL4/MatroidTopologicalBridge.lean
Sierpiński obstruction to exchange	sierpinski_not_topologicalClosureExchange	PEL4/MatroidTopologicalSierpinski.lean
Channel topology realizes channel closure	channelInteriorSemantics_closure_iff_channelClosure, channelInteriorSemantics_exchange	PEL4/MatroidTopologicalBridge.lean
Dependence iff member generated by remainder	channelDependent_iff_mem_closure_erase	PEL4/MatroidEvidenceCircuits.lean
Complete circuit classification	channelCircuit_iff_exists_samePolarity_pair	PEL4/MatroidEvidenceCircuits.lean
Circuit closure redundancy and elimination	channelClosure_pair_iff_singleton_of_samePolarity, channelPair_circuit_elimination	PEL4/MatroidEvidenceCircuits.lean
Contraction-style background loop effect	background_generates_samePolarity, background_does_not_generate_oppositePolarity	PEL4/MatroidEvidenceCircuits.lean
Explicit-ground deletion and contraction preserve matroid closure axioms	GroundMatroidClosure.delete, GroundMatroidClosure.contract	PEL4/MatroidEvidenceMinors.lean
Contraction loops equal closure of contracted set	GroundMatroidClosure.contract_isLoop_iff_mem_closure	PEL4/MatroidEvidenceMinors.lean
Finite channel contraction loop profile	finiteChannel_contractElem_isLoop_iff	PEL4/MatroidEvidenceMinors.lean
Parallel deletion preserves coarse FDE state	realizesFDE_delete_parallel_iff	PEL4/MatroidEvidenceMinors.lean
Unique deletion removes exactly one channel	delete_unique_channel_profile	PEL4/MatroidEvidenceMinorSemantics.lean

Canonical finite channel matroid is loopless	`finiteChannel_no_loops`, `finiteChannel_supportsNonloop_iff_supportsGround`	`PEL4/MatroidEvidenceMinorSemantics.lean`
Contraction removes nonloop capacity from exactly one channel	`finiteChannel_contract_sameChannel_no_nonloop`, `finiteChannel_contract_otherChannel_supportsNonloop_iff`	`PEL4/MatroidEvidenceMinorSemantics.lean`
Gap-free iff channel-complete; glut-free iff channel-consistent	`realizesFDE_gapFree_iff_channelComplete`, `realizesFDE_glutFree_iff_channelConsistent`	`PEL4/ClassicalRecovery.lean`
Complete and consistent evidence recovers exactly T/F	`realizesFDE_channelRegular_iff_classical`	`PEL4/ClassicalRecovery.lean`
Recovered T/F slice is closed under FDE connectives	`classicalValue_not`, `classicalValue_and`, `classicalValue_or`	`PEL4/ClassicalRecovery.lean`
Tolerant and strict excluded-middle recovery	`excludedMiddle_designated_iff_gapFree`, `excludedMiddle_eq_T_iff_classical`	`PEL4/ClassicalRecovery.lean`
Glut-freedom restores LP explosion	`glutFree_restores_LP_explosion`, `realizesFDE_channelConsistency_restores_LP_explosion`	`PEL4/ClassicalRecovery.lean`
Modal stability propagates recovered classicality	`stable_regular_value_is_classical_through_accessibility`	`PEL4/ClassicalRecovery.lean`
Parallel deletion preserves channel regularity	`channelRegular_delete_parallel_iff`	`PEL4/ClassicalRecovery.lean`
Unique deletion from regular evidence forces N	`delete_unique_from_channelRegular_not_complete`, `realizesFDE_delete_unique_from_regular_eq_N`	`PEL4/ClassicalRecovery.lean`
Propositional formulas preserve the recovered T/F slice	`eval_isClassical_of_propositional`	`PEL4/FormulaClassicalRecovery.lean`
Formula-level strict LEM and restricted LP explosion	`eval_excludedMiddle_eq_T_of_propositional`, `propositional_contradiction_LP_valid`	`PEL4/FormulaClassicalRecovery.lean`
Classical atoms can yield a Lockean belief gap	`gate6BeliefBoundary_atomicClassical`, `gate6BeliefBoundary_belief_is_N`	`PEL4/FormulaClassicalRecovery.lean`
Raw possibility and evidence-stable knowledge preserve classical profiles	`modalRawPossibilityValue_classical_of_profile`, `modalKnowledgeValue_classical_of_profile`	`PEL4/ModalFormulaClassicalRecovery.lean`
Every belief-free modal formula is classical under classical atoms	`evalModal_isClassical_of_beliefFree`	`PEL4/ModalFormulaClassicalRecovery.lean`
Belief is classical exactly at threshold regularity	`belief_isClassical_iff_thresholdRegular`	`PEL4/BeliefClassicalRecovery.lean`

Balanced belief witness is incomplete but consistent	`gate6BeliefBoundary_not_thresholdComplete`, `gate6BeliefBoundary_thresholdConsistent`	`PEL4/BeliefClassicalRecovery.lean`
Full legacy formulas recover classically under recursive admissibility	`eval_isClassical_of_recoveryAdmissible`, `eval_isClassical_of_global_recoveryAdmissible`	`PEL4/BeliefClassicalRecovery.lean`
Classical accessible profiles have complementary support masses	`classicalSupportMasses_sum_one`	`PEL4/LockeanThresholdPhaseTransition.lean`
Supermajority probability integrity derives threshold consistency	`probabilityIntegrity_classicalProfile_thresholdConsistent`	`PEL4/LockeanThresholdPhaseTransition.lean`
Under integrity, belief is classical iff threshold-complete	`probabilityIntegrity_classicalProfile_beliefClassical_iff_complete`	`PEL4/LockeanThresholdPhaseTransition.lean`
At-most-half thresholds exclude gaps; supermajority thresholds exclude gluts	`complementaryMasses_atMostHalf_noGap`, `complementaryMasses_supermajority_noGlut`	`PEL4/ThresholdPhaseGeometry.lean`
Threshold regularity reduces to consistency below/at one half and completeness above one half	`complementaryMasses_atMostHalf_regular_iff_consistent`, `complementaryMasses_supermajority_regular_iff_complete`	`PEL4/ThresholdPhaseGeometry.lean`
Exact 50/50 threshold phase dichotomy	`halfHalf_thresholdGlut_iff_atMostHalf`, `halfHalf_thresholdGap_iff_supermajority`, `halfHalf_phase_dichotomy`	`PEL4/ThresholdPhaseGeometry.lean`
No inclusive threshold regularizes the 50/50 tie	`halfHalf_never_thresholdRegular`	`PEL4/ThresholdPhaseGeometry.lean`
Probability-integrity formula recovery needs only belief completeness	`eval_isClassical_of_probabilityRecoveryAdmissible`, `eval_isClassical_of_global_probabilityRecoveryAdmissible`	`PEL4/ProbabilityIntegrityFormulaRecovery.lean`
Nonclassical belief is exactly incompleteness on classical integrity profiles	`probabilityIntegrity_classicalProfile_beliefNonclassical_iff_incomplete`	`PEL4/ProbabilityIntegrityFormulaRecovery.lean`
Split CPEL translations exactly recover both FDE support bits	`evalCPEL_translation_bits`	`PEL4/CPELClassicalCollapse.lean`
Two translated Boolean channels represent the full FDE value	`eval_eq_evalCPELPair`	`PEL4/CPELClassicalCollapse.lean`
Classical values are exactly the complement locus of the split channels	`classicalValue_iff_neg_eq_not_pos`, `isClassicalValue_iff_splitTranslations_complement`	`PEL4/CPELClassicalCollapse.lean`

Probability recovery collapses and reconstructs the split representation from one Boolean channel	`probabilityRecovery_splitTranslations_collapse`, `probabilityRecovery_eval_reconstructed_from_tr_pos`	`PEL4/CPELClassicalCollapse.lean`
Strict FDE truth is exactly the split Boolean profile $(\mathsf{true}, \mathsf{false})$	`eval_eq_T_iff_split_strict`	`PEL4/CPELConsequenceBridge.lean`
LP and ST consequence are exactly represented in the induced CPEL evaluator	`LP_SemanticEntails_iff_CPELPositiveSemanticEntails`, `ST_SemanticEntails_iff_CPELSplitSTSemanticEntails`	`PEL4/CPELConsequenceBridge.lean`
ST derivability is sound in the induced split CPEL semantics	`ST_derivation_sound_in_induced_CPEL`	`PEL4/CPELConsequenceBridge.lean`

Verification note. The full Lean 4.31 CI build and strict audits described in Section 11 passed. The audited manuscript and MPFG declarations have only standard Lean axiom dependencies. Their statements, not informal paraphrases, determine the precise scope of verification. The formula-level lift from fixed events to arbitrary probability-sensitive support events remains separate and is not inferred from the model-path theorem.

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